Diplomarbeit

Application of statistical analysis methods on ToF-SIMS data of native and degraded polymers

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Contents

1 Abstract

The widespread use of polymer coatings as passivation layers in the semiconductor industry has led to growing interest in understanding material and surface changes in technologically relevant polymers. One critical aspect in this context is the migration of ions and water through the polymer matrix, which can significantly compromise the long-term performance and reliability of semiconductor components, due to corrosion and other failure modes. As such, accurately tracking the progression of degradation within polymer layers is essential for evaluating their suitability in advanced electronic applications.

To gain insight into near-surface changes and monitor degradation profiles with high depth resolution, Time of Flight Secondary Ion Mass Spectrometry (ToF-SIMS) was employed. While ToF-SIMS is not traditionally used for polymer analysis, its ability to provide high depth resolution and surface sensitivity makes it a promising technique for this purpose. However, the method yields large volumes of complex spectral data, as Tof-SIMS records a full mass spectrum over a defined mass range for each pixel, repeated for every layer in a depth profile. This necessitates the use of chemometric tools for efficient and meaningful interpretation. Principal Component Analysis (PCA) and Hierarchical Cluster Analysis (HCA) were utilized to reduce dimensionality and identify key patterns in the data based on simple spectral descriptors.

Initial studies demonstrated that common polymers could be successfully differentiated using ToF-SIMS in combination with chemometric analysis. These findings were further validated and extended by investigating the degradation behavior of high-performance polyimide (PI) and poly(p-phenylene-2,6-benzobisoxazole) (PBO) films deposited on silicon wafers. Accelerated aging was achieved through controlled ultraviolet (UV) radiation exposure, creating a simplified model system for studying near-surface degradation. Subsequent depth profile analysis revealed qualitative differences in the aged samples, including estimation of UV penetration depth when comparing near-surface to bulk level signal.

The methodology developed in this work lays the groundwork for broader applications. In future experiments, more complex and industrially relevant degradation processes, such as water exposure, plasma treatment, and environmental aging, will be explored. The combination of ToF-SIMS and multivariate data analysis presents a robust framework for uncovering degradation mechanisms in polymer films, ultimately contributing to the development of more durable and reliable materials for microelectronic devices.

2 Kurzfassung

Der weit verbreitete Einsatz von Polymerbeschichtungen als Passivierungsschichten in der Halbleiterindustrie hat zu einem wachsenden Interesse an der Untersuchung von Material- und Oberflächenveränderungen in technologisch relevanten Polymeren geführt. Ein zentraler Aspekt ist dabei die Migration von Ionen und Wasser durch die Polymermatrix, die langfristig die Leistung und Zuverlässigkeit von Bauelementen durch Korrosion oder andere Zersetzungsprozesse beeinträchtigen kann. Um die oberflächennahe Degradation von Polymeren nachzuverfolgen und deren Eignung für moderne elektronische Anwendungen zu bewerten, ist eine Analyse mit hoher Tiefenauflösung erforderlich.

Zur Untersuchung solcher Veränderungen wurde die Flugzeit Sekundärionen Massenspektrometrie (ToF-SIMS) eingesetzt. Eine Methode, die zwar traditionell seltener in der Polymeranalytik Anwendung findet, aber aufgrund ihrer hohen Tiefenauflösung und Oberflächenempfindlichkeit großes Potential bietet. Da ToF-SIMS vollständige Massenspektren über einen definierten Massenbereich für jedes Pixel und jede Schicht eines Tiefenprofils aufzeichnet, fallen große Datenmengen an, deren effiziente Auswertung den Einsatz chemometrischer Methoden erfordert. Hierfür wurden Hauptkomponentenanalyse (PCA) und hierarchische Clusteranalyse (HCA) auf einfache spektrale Deskriptoren angewendet, um die Komplexität der Daten zu reduzieren und relevante Informationen sichtbar zu machen.

Erste Untersuchungen zeigten, dass gängige Polymere mit dieser Kombination aus ToF-SIMS und Chemometrie erfolgreich voneinander unterschieden werden können. Aufbauend auf diesen Ergebnissen wurde das Degradationsverhalten von hochleistungs Polyimid (PI) und poly(p-phenylene-2,6-benzobisoxazole) PBO-Filmen, abgeschieden auf Siliziumwafern, unter gezielter UV-Bestrahlung als Modellsystem für Alterungsprozesse untersucht. Die daraus resultierenden Tiefenprofile machten qualitative Unterschiede zwischen gealterten und frischen Proben deutlich und erlaubten Rückschlüsse auf die Eindringtiefe der UV-induzierten Degradation.

Die im Rahmen dieser Arbeit entwickelte Methodik bildet eine Grundlage für weiterführende Studien. In künftigen Experimenten sollen komplexere und industrienahe Degradationsprozesse, wie Feuchteeinwirkung, Plasmabehandlung oder Alterung durch Umwelteinflüsse, untersucht werden. Die Kombination aus ToF-SIMS und multivariater Datenanalyse bietet dabei einen vielversprechenden Ansatz zur Identifikation und Beschreibung von Degradationsmechanismen in Polymerfilmen und leistet somit einen Beitrag zur Entwicklung langlebiger und zuverlässiger Materialien für mikroelektronische Anwendungen.

3 Introduction

High end polymers are used on a regular basis by the semiconductor industry in a wide spectrum of applications, ranging from passivation layers, to high temperature adhesives and dielectric layers and many more. Their versatility and performance under extreme conditions, including high temperatures or mechanical stress, make them indispensable in the design and manufacturing of advanced microelectronic devices. As product lifetime and reliability is of ever greater importance, a focus on uncovering material and surface changes in certain used polymers have become of heightened interest, as even small changes in composition by degradation can affect performance. A major contribution for degradation is by migration of ions and water through the polymer, inducing corrosion and/or structural weakness over time, being of detriment to long term functionality of microelectronics.

To track progression of degradation through the polymer, surface sensitivity and high depth resolution is needed, as these types of degradation often occur near the surface and can advance without introducing new elements, but only functional changes of the polymer matrix. Conventional analysis and surface analysis methods often lack the combination of surface sensitivity, depth resolution and elemental and molecular information for tracking these changes. A more unconventional method in polymer analysis, namely Time of Flight Secondary Ion Mass Spectrometry (ToF-SIMS),[1] does uniquely provide all required features. Although traditionally seeing limited use in polymer analysis, the introduction of the Ar-cluster gun boosted its involvement in polymer analysis by quite a lot, enabling gentle sputtering of organic and biological samples in the life sciences, drastically reducing fragmentation of molecular ions.[2] ToF-SIMS produces large amounts of data while depth profiling, as each pixel contains an entire mass spectrum, with resolutions up to 2048x2048 pixel and mass ranges in polymers beyond 10,000 u, being repeated for each layer analyzed. To reduce the immense amount of data ToF-SIMS produces, established chemometric methods, such as PCA[3] and HCA, will be applied to simple spectral descriptors, effectively reducing data from entire mass spectra to a few selected characteristic secondary ions.

Tracking diffusion of extrinsic ions, such as potassium, which do not natively occur in the polymer is already well established,[4] but primarily focuses on the introduced ions and their diffusion behavior, like diffusion rate, rather than the changes within the polymer itself. However oxidation or water uptake do not necessarily introduce new elements into the sample and only manifest in shifted relative peak intensities in the mass spectrum, posing new challenging data evaluation. The core objective of this work is to differentiate between degraded and native polymer samples using chemometrics,[5] expanding the classifier to not only differentiate the polymer type,[6] but differentiate the “slices” acquired from depth profiling within the same sample. As compared to analyzing cross sections,[6] where lateral resolution of ToF-SIMS is the limiting factor, depth resolution far surpasses lateral resolution enabling better determination of degradation. Some aspects of this approach can be found in literature, but have focused on static surface characterization and classification, without characterizing depth profiles, therefore lacking in insights into the extent of degradation into the polymer.[7–9]

Initial experiments will be carried out using high-end commercial polymers and powdered polymers, to establish Tof-SIMS settings and method validation. Altering the samples will be done by UV radiation, as it is by far the simplest and fastest method to introduce degradation into the polymer, setting up a repeatable model system. Measurements will take place on the IONTOF ToF-SIMS 5, where suitable measurement parameters and compromises in settings need to be found, to support analysis with a focus on polymers. The developed data processing method can then be implemented in further practical experiments, using different, industry relevant degradation mechanisms.

4 Experimental Goals

This work will deal with the following points:

- **Establish measurement parameters to analyze polymers.** Measurement settings for ToF-SIMS need to be determined, to adequately measure polymers. Topics that need to be adressed for a reliable measurement outcome include induced electrostatic charge removal with an electron flood gun[10] or different raster methods, appropriate selection of ToF cycle time for determining the target mass range, determination of the primary ion species (Bi_1^+, Bi_3^+, Bi_5^+, Bi_7^+) influencing the creation of higher mass fragments and the signal intensity reaching the detector

- **Apply chemometrics to differentiate polymers by their ToF-SIMS spectra.** Train a classifier (PLS-DA, Random Forests, ...) or use other chemometric methods (HCA, PCA, ...) to differentiate between the following polymers: PI (polyimide), PS (polystyrene), PVP (polyvinylpyrrolidone), PMMA (polymethyl methacrylate), PVC (polyvinyl chloride)

- **Controlled degradation experiment with UV light.** Perform controlled UV-aging of PI and PBO (poly(p-phenylene-2,6-benzobisoxazole)) utilizing the knowledge acquired in the previous points.

- **Establish feasibility of differentiating degraded from unaltered polymer.** Test feasibility of differentiating ToF-SIMS spectra of UV-degraded polymer and unaltered polymer, using the same chemometric approach as in the last point. Following a positive outcome, the extent of degradation by comparing depth profiles of altered and non altered samples will be subsequently determined.

- **Develop data pipeline for chemometric evaluation.** Prepare data in SurfaceLab 7 and use Python to bring data into a useable form for statistical evaluation using scikit-learn, independent of the sample or degradation mechanism of interest.[11]

5 Background

5.1 ToF-SIMS

Time of Flight Secondary Ion Mass Spectrometry (ToF-SIMS) works, by bombarding a sample with a pulsed, focused primary ion beam which is rastered over its surface. The impacting primary ions generate a collision cascade near the surface, ejecting neutrals, molecules, secondary electrons and secondary ions alike. Different primary ions are available (Bi, Au, Ga, ...) and can be used to varying effects in applications throughout many industry sectors.[12] Generated secondary ions are extracted into the time of flight analyzer by applying an electrical potential (2000 V). To enable high transmission of ions and keep the surface free of adsorbates, the sample is analyzed under ultra high vacuum (10^{-8} - 10^{-10} mbar) conditions. The ToF-analyzer separates the different ion species by mass to charge ratio. A multichannel plate detector (MCP) detects and amplifies the signal from the incoming ions, by single ion counting. The uniqueness of ToF-SIMS lies in its extremely high sensitivity, enabling the detection of trace amounts of virtually every element, including their isotopes, across the periodic table, all with excellent mass and spatial resolution. In addition, the use of a second ion gun, known as the sputter gun, allows for controlled erosion of the sample surface, enabling not only surface level elemental and molecular analysis but also depth profiling from a few nanometers to several micrometers. Figure 1 shows a schematic overview of the individual components that make up the IONTOF ToF-SIMS 5. Each component will be introduced in the following sections.

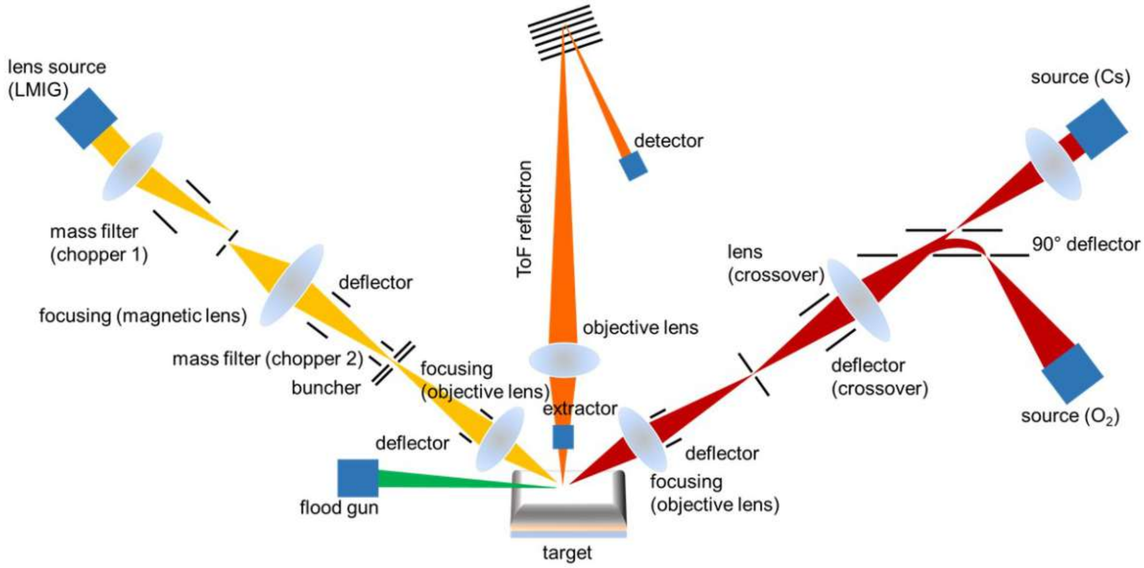


Figure 1: Schematic depiction of the different components of ToF-SIMS. Left: Primary ion source and its electrostatic lens setup, including a mass filter for selecting ion species. Middle: Time of flight mass analyzer including a reflectron for focusing ion energies. Right: Dual Source Column (DSC), secondary sputter ion gun, capable of generating Cs^+, O_2^+ and Ar^+ ion beams. Bottom: Electron flood gun used for charge compensation.

5.1.1 Primary Ion Gun

To generate primary ions a liquid metal ion gun is used, of which the IONTOF ToF-SIMS 5 is equipped with a Bi/Mn emitter. Ions are generated by melting the bismuth manganese alloy, wetting a tungsten needle, and applying an extraction voltage to the tip, which causes ions to accelerate off the emitter surface, via field ionization, into a region called Taylor cone (Figure 2).[13] Extracted ions are usually accelerated to a maximum of 25 keV and passed into a time of flight mass filter. The tube in which the LMIG gun is located also comprises a mass filter which enables the selection of ion species produced at the source by chopping the laterally spread out ion clouds. Different ionized clusters of bismuth (Bi_n^{p+}) and manganese all get accelerated in the electric field, and end up with the same kinetic energy, in this case 25 keV. By exploiting the different velocities that ion species develop due to different masses, the mass filter can be set to filter the spread out ion packets, via nanosecond timings, and select the desired species. The selected ion packet subsequently interacts with an ion buncher, which focuses the ion packets, by decelerating the front of the ion cloud via a repulsive electric field. Having all the ions of one pulse interact with the sample at ideally the exact same time is of high importance for mass separation via the reflectron time of flight mass analyzer, as even small differences in velocity/impact time influence mass resolution.

The choice of Bi over previous LMIG iterations, which used Au or Ga, is supported by several clear advantages. Bismuth has very high atomic mass (209 u), is monoisotopic, yet (technically) still stable,[14] easily forms clusters thereby achieving high impact mass and has a quite reasonable low melting point of 271.40 °C.[15]

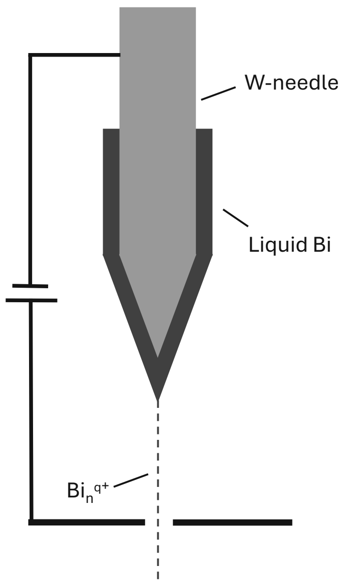


Figure 2: Liquid metal ion gun LMIG, ionizing liquid bismuth on tungsten needle by field ionization.

5.1.2 Dual Source Column (DSC)

Addition of a second, high current ion source to the instrument, allows depth profiling of samples. The main concept of this addition is to alternate between analyzing the sample surface with primary ions from the LMIG, and eroding of the topmost layers with the high current sputter gun from the DSC. This way, three-dimensional depth profiles can be compiled. As depth profiles are recorded as intensity against sputter time, additional steps to determine the sputter depth are required. To calculate sputter depth a sputter rate needs to be determined by measuring the depth of the eroded crater using profilometry or ellipsometry, alternatively sputter rates can be calculated using TRIM/SRIM simulations.[16] As sputter rate is dependent on the ion yield, which itself is highly dependent on the sampled matrix, this value needs to be individually calculated for chemically different layers, and/or samples.

The DSC of the IONTOF ToF-SIMS 5 model used for this work features three different ion beams. First an electron ionization O_2^+ ion gun, ideally used for measuring in positive mode. As the sample is eroded, O_2 is implanted into the sample and a process called atomic mixing takes place.[13] Due to the high electronegativity of oxygen, when sampling with the primary ion beam, the implanted oxygen tends to boost electron transitions from neutrals to oxygen, resulting in higher abundance of generated positive secondary ions.

As an alternative, Ar^+ can be used in this setup, also utilizing the electron ionization source. In specific circumstances, especially if one is interested in analyzing oxygen content, it is not of benefit to use ${O_2}^+$, as implantation can significantly alter chemical species or structure of a sample. Sputtering with argon allows to mitigate this effect, but still acquire viable intensities in positive mode.

For measuring in negative mode the Cs^+ emitter can be used, which relies on surface ionization for ion generation. Being on the very opposite end of the electronegativity scale, implanted Cs tends to raise the materials work function at the samples surface, leading to higher probability of electrons to leave the solid, increasing negative ion yield.[13]

In principle each sputter species can be used while measuring in negative or positive mode, but depending on the sample and the elemental/molecular information targeted, most of the time the choice is quite clear on which of the two main secondary beams (Cs^+, ${O_2}^+$) should be used, with the occasional outlier requiring Ar^+. For example alkali metals are easily ionized to Me^+, therefore being highly sensitive in positive mode. On the other hand halogens are very sensitive in negative mode, due to their tendency to form X^-. Often if an element is not very sensitive in either mode, adduct ions can be used for evaluation, for example N can hardly be detected in negative mode, but CN^- can.

5.1.3 Electron Flood Gun

An electron flood gun is implemented into the device, to mitigate electrostatic charges of the sample. During ion bombardment ions are introduced to the sample and secondary ions are emitted from the sample by the sputtering process. Additionally a high amount of secondary electrons is emitted. Due to the vast amount of ejected secondary electrons samples tend to acquire an overall positive charge, which can be corrected by flooding the sample in an electron shower introduced by the flood gun, if the material itself is not conductive enough to dissipate the charges.[13] Practically the floodgun is a heated filament ejecting electrons, often called a Wehnelt. Voltage between 0 and 21 V can be applied to the filament between and current of up to 2.5 A.

5.1.4 Time of Flight Mass Analyzer

To separate and analyze the produced secondary ions, a mass filter is used, as previously mentioned. As the secondary ions are generated at the sample surface, an electric potential draws them into the ToF. Secondary ion yields are usually quite low and span many orders of magnitude, i.e., from 10^{-7}–1%. The transmission into the ToF is not stated by the manufacturer, but can theoretically under ideal conditions reach 90-100%.[13] All secondary ions are accelerated by the electric field to the same kinetic energy $E = \frac{mv^2}{2} = Uq$, resulting in different velocities proportional to their mass. Due to different travel time through the ToF tube, different mass to charge ratios arrive at slightly different times at the detector. To achieve high mass resolution with this analyzer, it is imperative for secondary ions to be generated in as little time frame as possible, ideally simultaneously, as to not influence ion travel time due to delayed generation/extraction. This is achieved by pulsing the primary beam, creating individual, highly timed events for secondary ion generation. To further improve upon the pulse duration of primary ions, which are in the range of 10-20 ns of length, an ion buncher is deployed at the end of the primary ion column, to compress the ion pulse to sub-nanosecond times. The ion buncher applies an electric field to decelerate the front of an ion pulse, so ions further back in the pulse can catch up to the front of the ion packet. The buncher delay must be set appropriately to focus the ion packet onto the sample surface, to achieve near simultaneous impact of the packet.

Further improvements to mass resolution are obtained by the use of a reflectron ToF. Although all secondary ions are accelerated in the same electric field, ions of the same m/z still show a distribution in kinetic energy of a few eV. To reduce energy dispersion, the ion mirror in the ToF tube not only reverses the ion flight direction and increases flight time, but also compensates for kinetic energy differences. Faster ions of a given m/z penetrate deeper into the ion mirror's electric field and thus travel a longer path, while slower ions are reflected earlier. This path length compensation results in a narrower temporal spread, improving mass resolution by ensuring that ions of the same species arrive at the detector nearly simultaneously. As Equation 1 shows, a reflectron improves both parameters for resolution in a ToF mass analyzer.

$$R = \frac{m}{\Delta m} = \frac{t}{2\Delta t} \tag{1}$$

5.1.5 Static SIMS

In static SIMS only the surface is sampled without the use of the sputter gun. The definition of static SIMS is widely agreed upon to be an ion dose below 10^{13} ions/cm^2, but can change for different materials.[17,18] By keeping the ion dose low, incoming ions can be assumed to always hit a new, fresh spot on the surface, and not an already sampled and altered area. To calculate ion dose, some measurement parameters are needed, such as sampled area, pixel, cycle time and ion current. Ion current usually is measured during the startup process of the device to ensure stable and consistent operating conditions. All others are chosen by the operator specifically for the measurement task at hand. Typical values are 100x100 µm area, with 128x128 pixel and a cycle time of 50 µs, with ${Bi_1}^+$ current being around 2.30 pA.

$$\text{Ions/cm}^2 = \frac{I \cdot t}{z \cdot e \cdot A} \tag{2}$$

I .... beam current

t .... time for one frame

z .... ion charge

e .... elementary charge $1.60217 \cdot 10^{-19}$

A .... Area

$$\text{Bi}_1{}^+{}_{2.30\ \text{pA}} : \frac{\text{pixel} \cdot \text{cycle time} \cdot \text{beam flux}}{\text{area}} = 1.176 \cdot 10^{11}\ \text{ions}/cm^2 \tag{3}$$

$$\text{Bi}_3{}^+{}_{0.82\ \text{pA}} : \frac{\text{pixel} \cdot \text{cycle time} \cdot \text{beam flux}}{\text{area}} = 4.193 \cdot 10^{10}\ \text{ions}/cm^2 \tag{4}$$

This implicates that around 85 seconds, or in other words around 104 frames of continuous sampling, can be performed before reaching the static SIMS limit. With ${Bi_3}^+$ dose is even smaller, as ${Bi_3}^+$ is less abundant in the generated primary ions, sampling for 239 seconds, or 291 frames is possible. All 2D recordings in this work were below that threshold at around 60 frames.

5.1.6 Dynamic SIMS

The sputter beam has much higher ion flux by 4-5 orders of magnitude and therefore exceeds the static limit (Equation 5) with 1 keV on an area of 300x300 µm and a beam current of 80 nA, eroding the surface in each sputter cycle. Pixel count is not needed this time, as pixel and cycling time only calculate time for one frame. In all 3D settings 1 s sputter time was chosen, eliminating this step.

$$\frac{I \cdot t}{z \cdot e \cdot A} = 5.548 \cdot 10^{14}\ \text{ions}/cm^2 \tag{5}$$

5.2 Chemometric Methods

Chemometrics is an interdisciplinary field combining chemistry, mathematics, statistics and computer science with the aim of extracting useful information from large, complex, high dimensional chemical data sets. It relies on techniques from multivariate statistics, or machine learning to perform multivariate calibration, clustering, pattern recognition, classification and many more.[19]

5.2.1 Principal Component Analysis, (PCA)

The base concept of PCA is that in a multidimensional data space information is only present in directions where variance is not zero. By finding the direction with most variance, information can be concentrated. The direction with the highest variance is called principal component 1, PC1, and projected onto a line. Then the process is repeated and the direction with the most variance perpendicular to PC1 is picked and called PC2. The algorithm is repeated, until as many principal components as there are dimension in the initial data space, are found and all PCs being perpendicular to the ones before them. The first PCs tend to be most important as the overwhelming majority of information is condensed in them. The later PCs in the sequence tend to accumulate noise and are usually not significant and are, thus, disregarded.

This decomposition of the data space is fundamentally linked to the eigenvectors and eigenvalues of the covariance matrix, which correspond with the directions with the most variance, with the eigenvalues being proportional to information content of a certain direction. The initial data matrix is split into two matrices U and V, where U and V are orthogonal. V is called the loading matrix and describes the directions of the PCs and U is called the score matrix, describing the coordinates in the new rotated space.

$$X = U \cdot V^T \tag{6}$$

X is the data matrix and multiplication of U and V results in X', which approximates the original data matrix X, as PC's with low information content can be discarded. The scores are the new coordinates in the rotated space and loadings the weights applied to the original variables to give the scores. This enables exploratory data analysis, where a complex multidimensional dataspace can be projected onto a plane, presenting the most amount of information.[20]

5.2.2 Hierarchical Cluster Analysis (HCA)

In agglomerative HCA a multidimensional data space is analyzed with the goal of separating data into clusters of similar properties depending on the context of the used dataset. The algorithm starts by calculating the distances between all data points and picks the two data points closest to each other. Distances can be calculated using different methods, such as the more common Euclidian distance or Manhattan distance, or for special cases methods such as the Mahalanobis distance.[21] After the two closest data points are found, they are merged into one new data point, a cluster. Again, application of various methods, such as single linkage, average linkage, Ward's method, are possible, each providing different benefits for different data sets. This first acquired distance is then used to start construction of a dendrogram plot by which HCA is usually visualized. The two data points are placed next to each other on one axis and linked at the calculated distance on the other axis, different preferences on which axis to use pertain. The algorithm is then repeated, until all data points are merged and linked in the dendrogram, allowing for interpretation of the emerging clusters.[22]

6 Instrumentation, Chemicals & Samples

6.1 Chemicals

Polymers:

- PVP, polyvinylpyrrolidone, average Mw 1,300,000 by LS, Merck KGaA (Darmstadt, Germany)
- PVC, poly(vinyl chloride), carboxylated, aver. M.W.220000, aver. carboxyl content 1.8 wt.%, ThermoFisher (Kandel) GmbH (Kandel, Germany)
- PI, polyimide, 13 µm, Dupont™ Kapton® HN PI, Goodfellow GmbH (Germany)
- PS, polystyrene foil, 30 µm, Goodfellow GmbH (Germany)
- PMMA, polymethyl methacrylate, beads, average M.W. 35,000 ThermoFisher (Kandel) GmbH (Kandel, Germany)
- PI, polyimide coated Si-wafer, 11 µm film, 1 cm^2, Infineon Technologies AG (Germany)
- PI, polyimide foil, 13 µm, Dupont™ Kapton® HN PI, Goodfellow GmbH (Germany)
- PI, polyimide powder, P84 Polyimide powder 200 mesh STD, Ensinger Sintimid GmbH (Austria)

Solvents:

- Cyclohexanone, ACS reagent, ≥99.0%
- Ethanol absolute, 99.8+ % for analysis, ThermoFisher (Kandel) GmbH (Kandel, Germany)
- Anisole, 99 %, Extra Dry, AcroSeal™, ThermoFisher (Kandel) GmbH (Kandel, Germany)
- N-Methyl-2-pyrrolidon, biotech. grade, ≥99.7 %, Merck KGaA (Darmstadt, Germany)
- Anisole, 99% , Merck KGaA (Darmstadt, Germany)

6.2 Samples and Preparation

- PI, (Polyimide) coated wafers, 11 µm film, 1 cm^2, Infineon Technologies AG (Germany) kindly provided by Infineon Technologies AG, were used without further sample preparation. The exact com- position is not provided, including a proprietary mix of polyimide and performance enhancing additives

- PBO, poly(*p*-phenylene-2,6-benzobisoxazole) coated wafers, 10 µm film, 1 cm^2, kindly provided by Infineon Technologies AG, were used without further sample preparation. The exact com- position is not provided, including a proprietary mix of PBO and performance enhancing additives

- Si wafer, 200 nm SiO_2, Infineon Technologies AG (Germany), kindly provided by Infineon Technologies AG, were used to apply polymer solutions or fix polymer foils ontop.

- **Foil samples** PI and PS foil were cut to approximately 1x1 cm and fixed directly to individual silicon wafers with conductive carbon tape.

- **Drop cast samples:** PVC was dissolved in cyclohexanone (15 mg PVC/ml cyclohexanone), PVP was dis- solved in ethanol (60 mg PVP/ml ethanol) and PMMA was dissolved in anisole[23] (50 mg PMMA/ml anisole), PI powder was dissolved in N-Methyl-2-pyrrolidon (NMP) (10 mg PI/ml). All solutions were vortexed and finally sonicated for 10 min. 75 l of the liquid samples were drop cast each onto individual Si wafers and left to dry on a 60 °C hot plate for 1 h and then for 24 h at room temperature in a sample container.

6.3 Instrumentation

The instruments used for in work include:

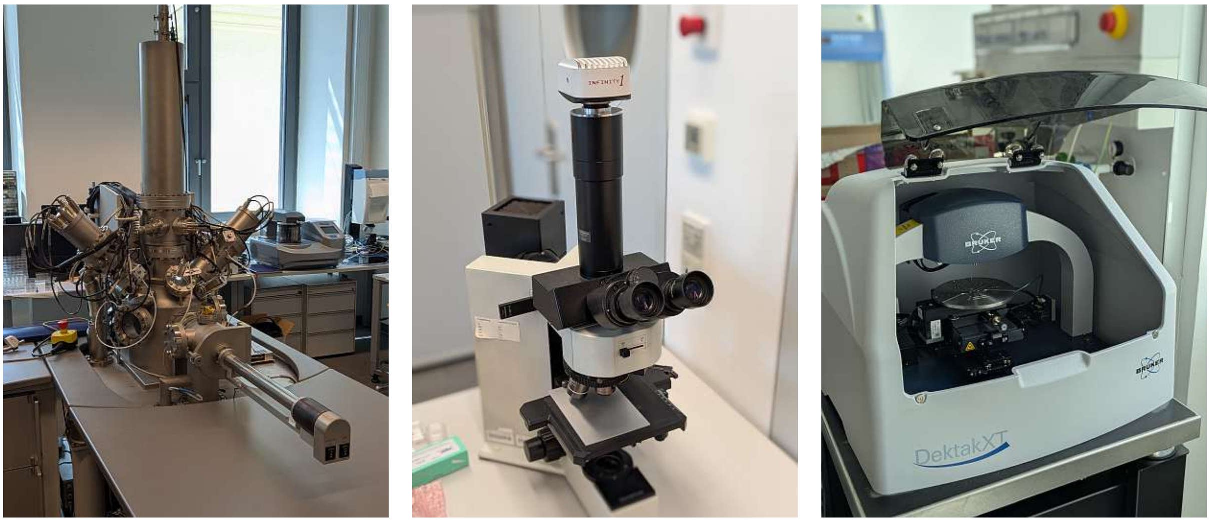
- IONTOF ToF-SIMS (IONTOF GmbH, Münster, Germany)

- Olympus BX60 optical Microscope (Olympus Austria GesmbH, Vienna, Austria)

- Bruker DektakXT (Bruker Austria GmbH, Vienna, Austria)

- Hot plate, Heidolph MR Hei-Standard (Heidolph Instruments GmbH & Co. KG, Schwabach, Germany)

- 11 W UV box



(a) (b) (c)

Figure 3: Used Instruments: a) IONTOF ToF-SIMS 5, b) Olympus Microscope, c) Bruker DektakXT.

7 Execution/Practical Part

7.1 Optimizing Measurement Parameters

When analyzing polymers, there are certain peculiarities and trade-offs to consider in ToF-SIMS, when compared to the usual inorganic, conductive samples. The following sections will detail the process of identifying and optimizing the measurement parameters.

7.1.1 Choosing the Primary Bi-Species

The Bi/Mn LMIG emitter is can produce a multitude of different ion species: ${Bi_1}^+$, ${Bi_2}^+$, ${Bi_3}^+$, ${Bi_4}^+$, ${Bi_5}^+$, ${Bi_6}^+$, ${Bi_7}^+$, ${Bi_1}^{2+}$, ${Bi_3}^{2+}$, ${Bi_5}^{2+}$, ${Bi_7}^{2+}$, Mn^{2+}
As described in 5.1.1 the chopper timings of the LMIG mass filter can be set to select the desired ion species. To determine the suitable ion species for polymer measurements, the list was reduced by exclusion of unsuitable ions first. Mn^{2+} was excluded, for its low abundance in the produced ion beam, and lacking the high mass and cluster forming ability of Bi. Additionally all doubly charged Bi ions were omitted due to low abundance. To choose from the single charged ion clusters, all even cluster indices were removed, as stability of clusters depends on the amount of valence electrons, with an even number of valence electrons having greater stability, hence more abundance among the formed primary ions. Bismuth has an odd number of valence electrons, therefore a cluster of an odd number of Bi atoms has an odd number of valence electrons, and by ionizing to a single positive charge, resulting in a cluster with an even amount of electrons. Bi cluster

with Bi_{2n+1}^+ are more stable than Bi_{2n}^+, leaving the odd numbered clusters to choose from.[24,25] As seen in Figure 4 higher cluster indices, comes lower abundance, and less signal in ToF-SIMS analysis, due to less ion current.[26] The final choice is mainly between Bi_3^+ and Bi_5^+. Bi_3^+ has excellent signal generation on the polymers in question, but depending on the state of the sample excessively high signals can saturate the detector, but can be mitigated in a few simple ways, which will be discussed later on. Bi_1^+ has less signal intensity, but as it is the smallest and lightest Bi-cluster species, mass fragments above m/z=50 become less intense in relation to smaller fragments.

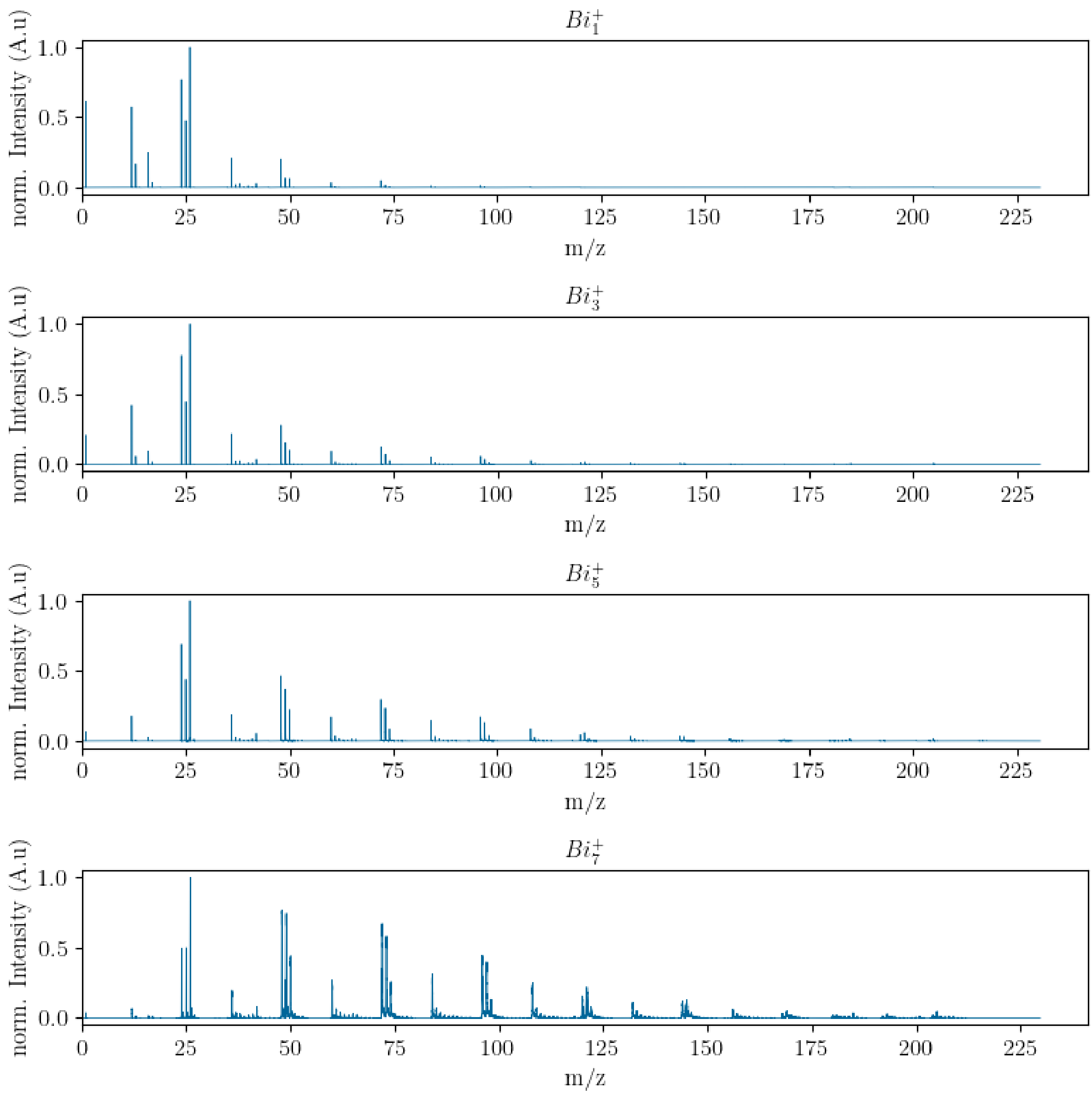


Figure 4: Normalized (m/z=26) ToF-SIMS spectra of polyimide foil, demonstrating the difference between Bi_1^+, Bi_3^+, Bi_5^+, Bi_7^+ clusters in signal intensity and higher mass fragment abundance. Sputtering for better counting rates, Cs 1 keV, 25 keV primary beam, negative mode, 50 µs cycle time, 128x128 pixel, non-interlaced 1-1-2, 75 s sputtering.

Bi_3^+ was chosen, as it offers a great compromise between signal intensity and fragmentation. Signal intensities is higher than with Bi_1^+, additionally offering more favorable fragmentation. Compared to Bi_5^+, Bi_3^+ total counts are decisively higher (Figure 5), and fragments concentrate more in the region up to around 50 m/z, where elucidation is still viable.

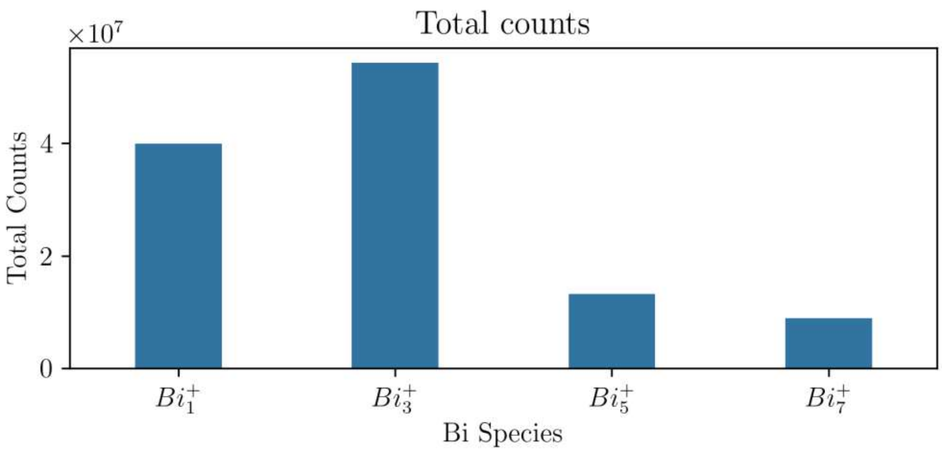


Figure 5: Comparison of the total counts of the odd numbered Bi-species, 3D for better counting rates, Cs 1 keV, 25 keV primary beam, negative mode, 50 µs cycle time, 128x128 pixel, non-interlaced 1-1-2, 75 s sputtering.

7.1.2 Cycle Time/Mass Range

For fast measurement times the cycle time of the ToF analyzer usually is set to 50 µs, as in inorganic analysis most of the periodic table is covered by this range, as 50 µs cycle time correspond to a m/z range between 0-230 u. For polymers cycle time needs to be adjusted, as m/z of macromolecules reach far beyond 230 u. To determine whether there is the necessity of going above m/z 230, spectra were recorded with the cycle time set to 105 µs (0-1016 u) and 330 µs (10043 u), see Figure 6.
Spectra recorded with 330 µs cycle time show no significant peaks beyond ≈500 u. Supporting this observation are the spectra recorded with 105 µs cycle, again not showing significant detection of ion species beyond 500 u apart from noise.

As data processing will be done using only lower mass fragments, and to speed up the measurement time, 50 µs were chosen for further measurements. Heavier fragments would not be useful, as their elucidation is not possible on ToF-SIMS, due to the fact that an unmanageable amount of combinations from C, H and N can be added up for each peaks m/z.

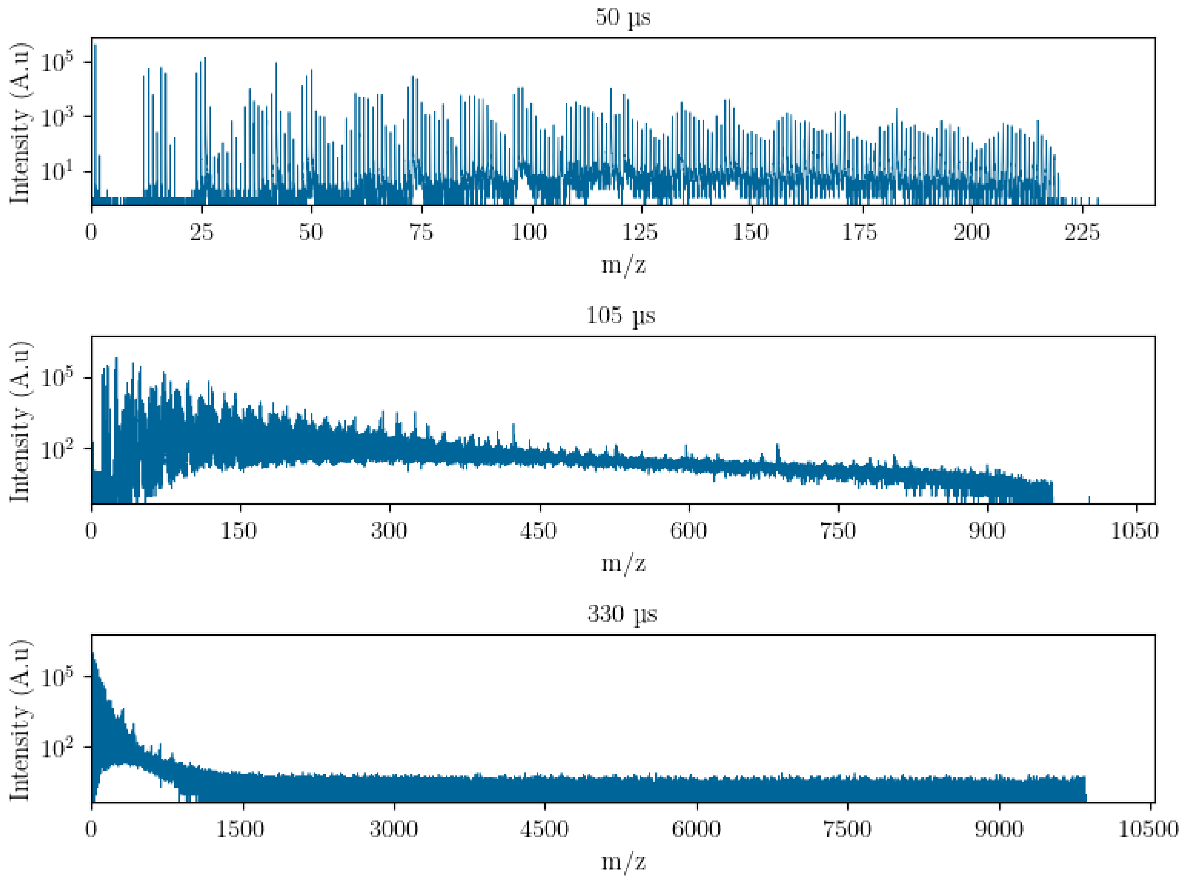


Figure 6: Comparison of different cycle times 50 µs 230 u, 105 µs 1016 u and 330 µs 10043 u, beyond 500 u only noise is observed. Cs 1 keV, Bi_3, negative mode.

7.1.3 Sputter Species

Three ion species can be produced by the DSC (Cs^+, O_2^+, Ar^+) as discussed in 5.1.2. Six permutations of ion species and ion mode (negative/positive) are possible, and were all tested for viability. The least effective is positive mode using Cs^+ and negative mode with O_2^+ as very low intensities were recorded as expected. Ar^+ did not live up to the performance of either O_2^+ in positive mode or Cs^+ in negative mode, and was not further considered, as its uses are very specific and it did not fit with the samples properties. See Figure 7 for a comparison of ion species performance. The choice fell onto Cs^+ in negative mode, as characteristic fragments are easier to determine in negative mode, as also supported by published data[27] As later on only smaller fragments were used for data processing, negative mode continued to be the choice, due to easier interpretable spectra, which include highly sensitive halogens in negative mode. If certain ions which are sensitive in positive mode are of interest, naturally positive mode with O_2^+ could be used.

Usually, 2 keV sputter energy gives the best results for sputtering, enabling highest focus and fastest sputtering, while 500 eV runs into focus issues, as space charge effects during slower and longer travel disperses ions in flight. 1 keV was chosen as a compromise of focus and sputter rate, with energy preferably being lower, as to not fragment the polymer even further during sputtering, and more importantly increase depth resolution due to slower sputtering.

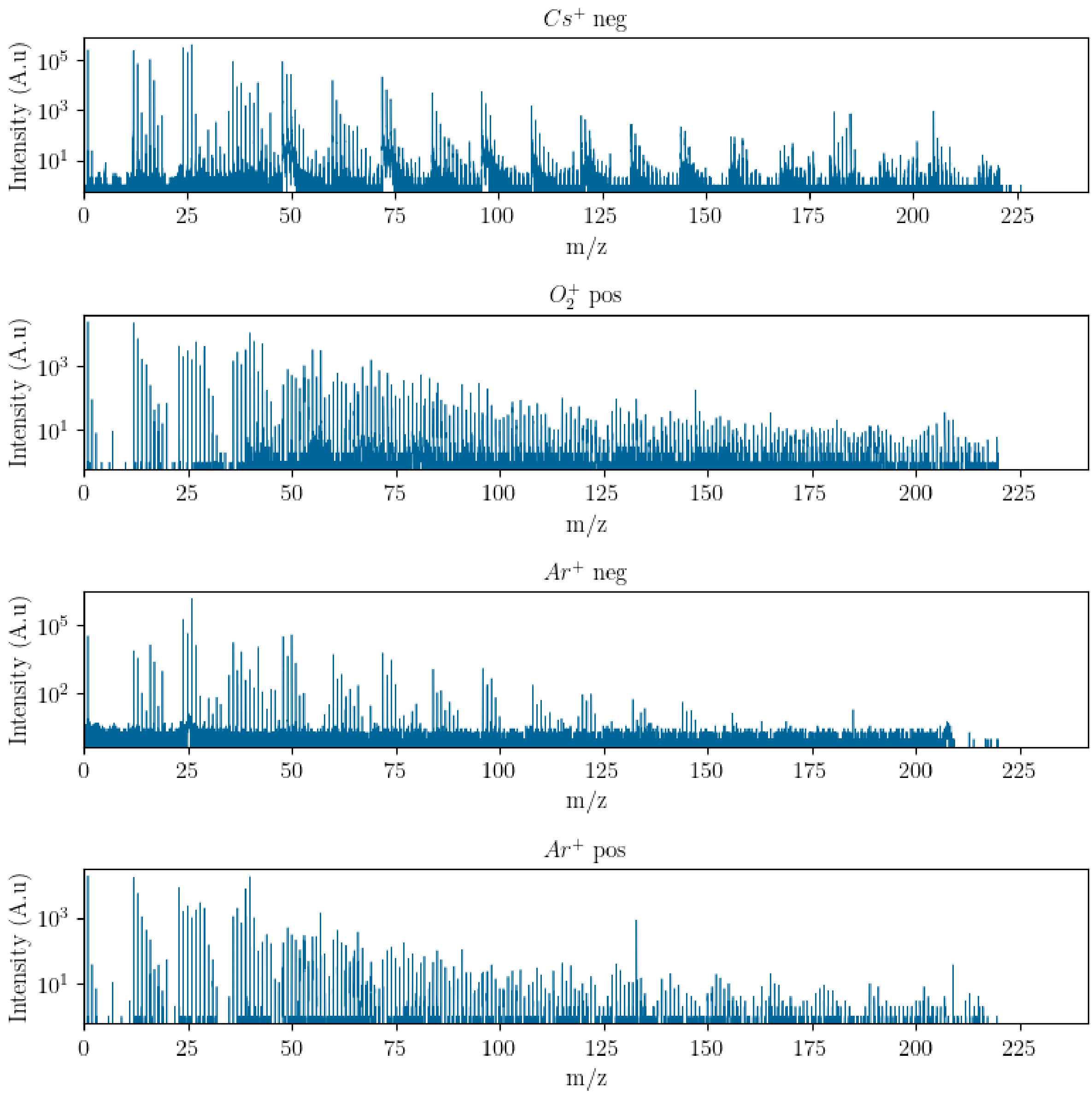


Figure 7: Comparison of all available sputter species Cs^+, ${O_2}^+$, Ar^+, using ${Bi_1}^+$ as primary species. Negative mode has 2 orders of magnitude more counts, and Cs^+ negative has more fragments than Ar^+ negative, 1 keV, 75 s sputtering, 1-1-2.

Ar^+ negative has massive overabundance of m/z=26 (CN^-) when compared to all of its other fragments. Ar^+ positive generally has the lowest signals of all chosen sputter species. ${O_2}^+$ in positive mode has very high C^+ signal compared to the next highest peaks, but generally intensities drop very fast over time during sputtering. The best fragmentation pattern for the analyzed polymer (PI) is achieved by using Cs^+ in negative mode, boasting around 10 times higher signal than the other species. Ar^+ positive has an artifact at m/z=137 originating from Cs contamination from a previous measurement using Cs^+, suggesting that in this case the distance between the sampled spots was not sufficient.

To determine the sputter rate in a given sample, a crater is eroded and afterwards depth is determined via profilometry. From the calculated sputter rate in nm/s recorded depth profiles can now be displayed with nm on the x-axis instead of sputter seconds. Already a higher sputter rate can be observed for the aged sample, when compared to the blank sample.

7.1.4 Charge Compensation

The standard flood gun settings of 20 V and 2.4 A provide electron energy that is sufficient to damage organic molecules, however since having more, or smaller molecular fragments is of no concern for the experiments in this work. Experiments with lower Wehnelt voltage, showed drastically worse performance, specifically in local charging effects of the polymer sample, distorting the ion beam, half the total counts and leading to malformed peaks in the mass spectrum. (As seen in Figure 8) On some thin polymer films, such as PMMA ($\approx$ 120 nm), discoloration could be observed after only a few seconds at 20 V. Reduction in voltage did not lead to improvements, so the floodgun settings were kept at the usual standard settings of 20 V and 2.4 A.

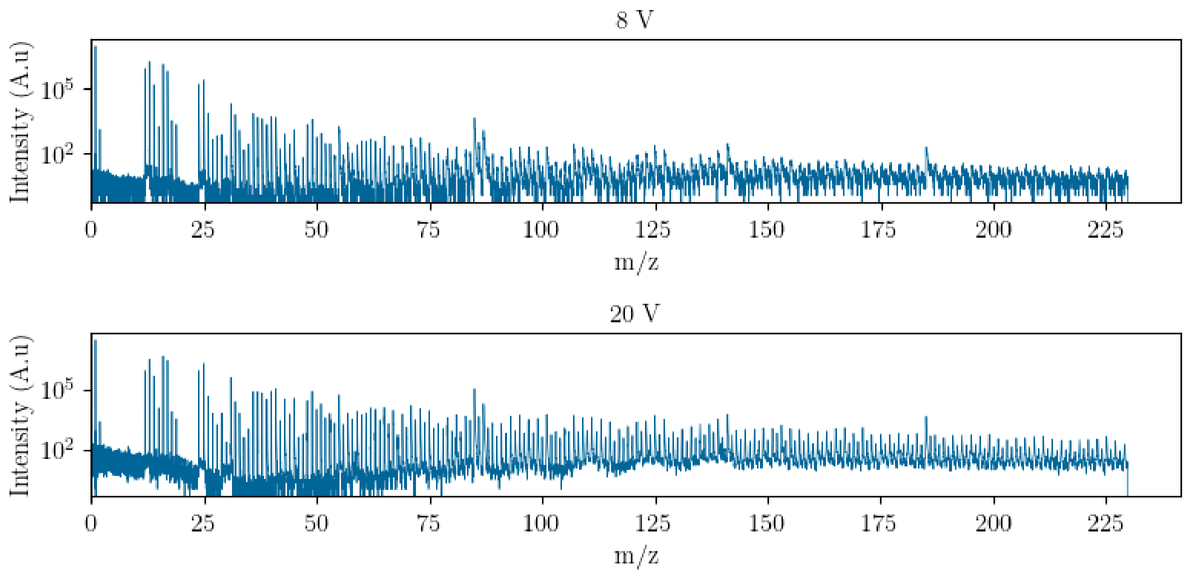


Figure 8: Comparing floodgun at minumum voltage 8 V and maximum voltage 21 V on a PMMA film, total counts for 8 V: $24 \cdot 10^6$, total counts for 21 V: $51 \cdot 10^6$.

In addition to the charge compensation achieved by the floodgun alone, the measurement mode can be changed between the standard interlaced and non-interlaced mode. In interlaced mode, the analysis beam and the sputter beam are triggered alternately. In non-interlaced mode, the analysis beam, sputter beam and charge compensation are triggered sequentially. The individual times can be customized, for example non-interlaced 1-1-1 means that one frame is captured, followed by one second of sputtering, followed by one second of only charge compensation. In general, non-interlaced mode was used in this work, as PI is highly insulating, requiring more charge compensation than the standard interlaced mode can provide. 1-1-2 did not show any additional improvements, and was only used on some PI foil samples, where charge compensation was especially hard, due to the foil being fixed to a wafer using conductive carbon tape.

By default, a sawtooth pattern is applied to the primary ion beam to raster over the analysis area, which may lead to undesirable charging. Further improvements can be achieved by using the random raster mode during primary ion frame acquisition, as it spreads out the sampled spots within the analyzed area.

7.1.5 Intensity

To compare mass spectra of different samples or even different sampling positions on the same sample, some form of signal normalization needs to be employed. Especially in ToF-SIMS matrix effects have a high impact on signal intensities leading to large intensity differences even in similar samples. The preferred method to normalize values is by dividing the measured species by the total counts acquired. Special care needs to be taken not to exceed the linear range of the detector, as saturated detector values complicate data processing. The instrument functions in single ion counting mode, encountering the usual issues surrounding detector dead time. High ion intensities lead to higher probability in two or more ions reaching the detector during a single counting cycle, effectively under representing the amount of counted ions, the detector is now saturated. This critical value can be calculated by the amount of pixels used for each frame ($128 \cdot 128$ pixel $=$ 16384 counts/frame) As total counts are used for normalization, if even one ion species drifts into saturation, all normalized values are now being misrepresented hence dividing by a value that can not be determined with certainty. SurfaceLab 7 employs dead time correction in the form of a Poisson correction to extend the linear range by a factor of 4, up to 65536 counts/frame, but even then counts of polymer samples can reach far above the corrected threshold.[28]

$$I_{corr} = -N \ln{(1 - \frac{I}{N})} \tag{7}$$

A few methods to reduce ion intensity can be applied. First primary ion pulse duration of, e.g. 17 ns with Bi_3^+, can be shortened in the LMIG ToF by changing the chopper timings. Shorter primary ion pulses means less ions reach the sample and subsequently the detector, reducing intensity. Secondly LMIG emission current can be reduced from the standard 1.00 µA to send even fewer ions to the sample. Finally less abundant primary ion species can be selected, for example Bi_5^+ over Bi_3^+.

To find an adequate compromise between not saturating the detector and still having high counting rate for data processing, it is necessary for each type of sample to first check the intensities in dynamic SIMS mode, as sputtering greatly changes the intensities and adjust the parameters accordingly, with chopper timings being the fastest and easiest, with the least amount of possible complications.

7.1.6 ToF-SIMS Settings

All findings regarding the ideal measurement settings, as established in the previous section, are compiled in Table 1. These settings will serve as the standard for subsequent experiments, unless otherwise specified.

Table 1: List of all major settings used for most measurements in this work, if not stated otherwise.

Primary Beam	
Ion Species	Bi_3^+
Beam Energy	25 keV
Emission current	1 µA
Pulse Width	13-16 ns
Analysis Area	100x100 µm
Raster Mode	Random
Resolution	128x128 pixel
Analyzer	
Extraction Voltage	2000 V
Extractor Mode	Negative
Cycle time	50 µs (0-230 u)
Measurement Mode	Non-Interlaced 1-1-1
Sputter Beam	
Ion Species	Cs^+
Beam Energy	1 keV
Sputter Area	300x300 µm
Flood Gun	
Energy	20 eV
Sample Chamber	
Pressure	$\approx 10^{-9}$mbar
Temperature	ambient

7.1.7 Measurements of Polymers

Using the established ToF-SIMS settings, measurements of 5 different polymers were performed, as shown in Figure 9. Although the polymers differ significantly in composition and structure, all could be successfully measured using the established settings. The resulting spectra are visually distinct, although some polymers exhibit similar features. Notably, PI and PS display a regularly repeating pattern above m/z=50. In almost all spectra, with the exception of PMMA, the distinctly most abundant ions are below m/z=40. The most abundant peak in all spectra is the H^- peak, which can be explained by the fact that static SIMS was used to record the spectra, sampling only the surface and its adsorbates.

7.1.8 Spectral Descriptors

Simple, characteristic fragments were chosen from the measured polymers, such as H^-, C^-, CN^-, Cl^-, F^-, C_2H^-, as most are present in some form in all tested polymers. The reason for not adding more complex fragments is simply the impracticability of first assigning the correct fragments to their peaks in the spectrum, which would require lengthy fragment structure elucidation for each individual polymer type using MS-MS methods such as QqQ or orbitrap. Adding the whole dataset is not feasible, as it is way too large and the focus should be placed on observing the change in small, known fragments. Approaches using the entire spectrum can be found, but offer no additional benefit for this work, as not being albe to assign the peaks in a spectrum does not result in interpretable results.[29]

Cl^- and F^- were chosen in all measurements, as they have very high sensitivity in negative mode ToF-SIMS, even if only present at a trace level. Due to differing matrices in the aged and native polymer, matrix effects can lead to changes in ion intensities as a result of varying ionization probabilities, or work functions. This effect would be pronounced in these very sensitive ions.

7.2 Differentiating Polymers with ToF-SIMS

To establish a solid experimental foundation for more complex studies and to demonstrate methodological feasibility, a simple classification of polymers was first conducted. The polymer samples used for the experiment are PI foil, PS foil, drop cast PVP, drop cast PMMA and drop cast PVC. Samples were measured with the previously established ToF-SIMS parameters, each sample was probed in 5 different locations using static SIMS. Characteristic ion peaks were selected in SurfaceLab 7, peak areas calculated and then

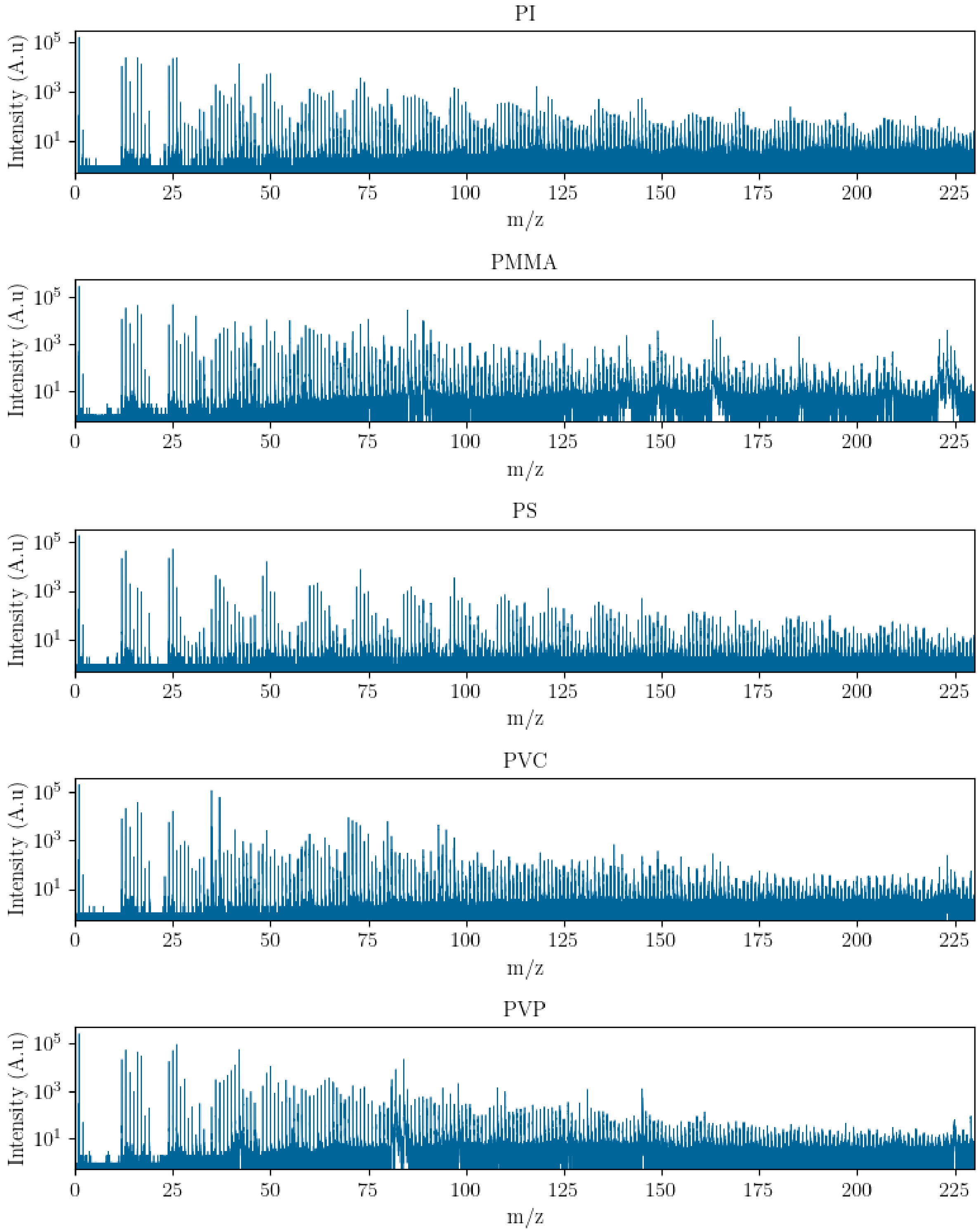


Figure 9: Comaparison of 5 polymers PI, PMMA, PS, PVC, PVP. Qualitative differences between spectra can be visually identified. Static SIMS, Bi_3^+, 60 frames.

exported. The exported files contain intensities of all chosen ions over the entire analysis time. Statistical data analysis was performed using Python scikit-learn package, where all individually exported files were added to one data set. Hierarchical cluster analysis, Figure 10, shows branching into 5 separate categories, which can be interpreted as the following polymer subcategories "contains nitrogen" for PI and PVP, "contains heteroatoms" for PMMA and PVC, and "contains only C and H" for PS.

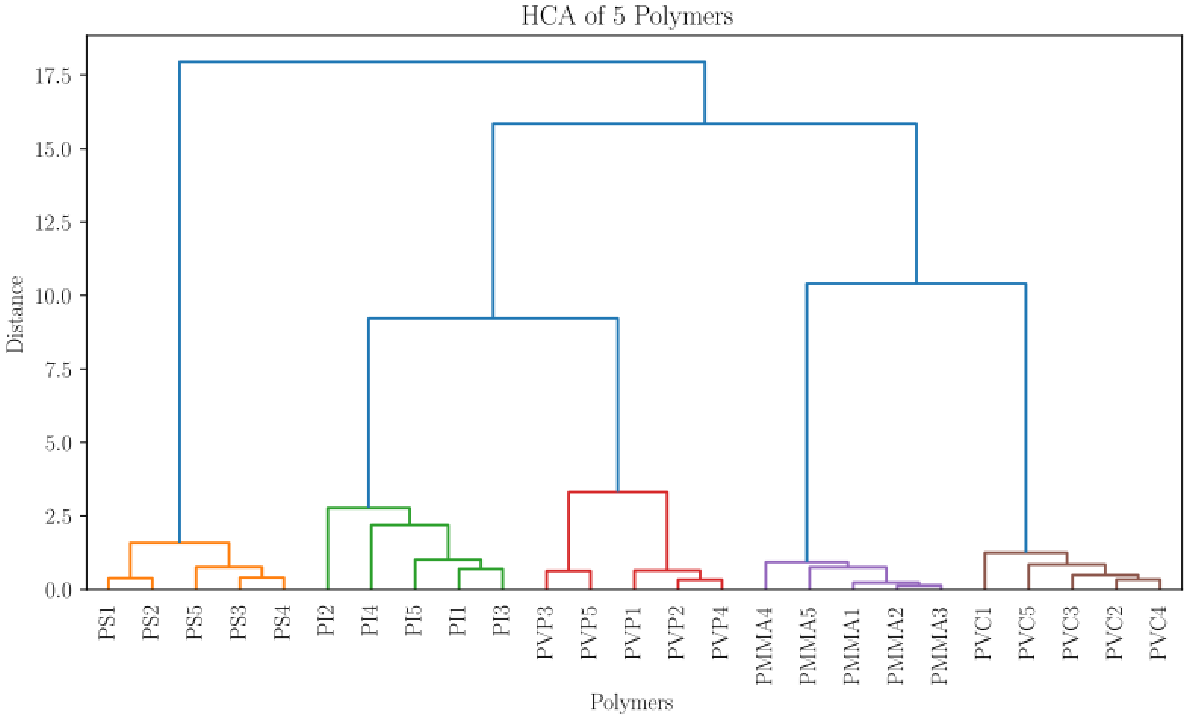


Figure 10: Hierarchical cluster analysis (HCA) of 5 measurements from 5 different polymers (PVP, PVC, PI, PS, PMMA). Normalized and standardized peak area of selected spectral descriptors, Ward's method, Euclidian distance.

Additional exploratory data analysis was performed using PCA. The same dataset was subjected to PCA, resulting in Figure 11. Loadings were plotted for interpretation of PC1 and PC2. Again, the separation is mainly due to heteroatoms, as PS is most clearly separated on the PC1 axis. PC2 then splits the heteroatoms between nitrogen and chlorine/oxygen.

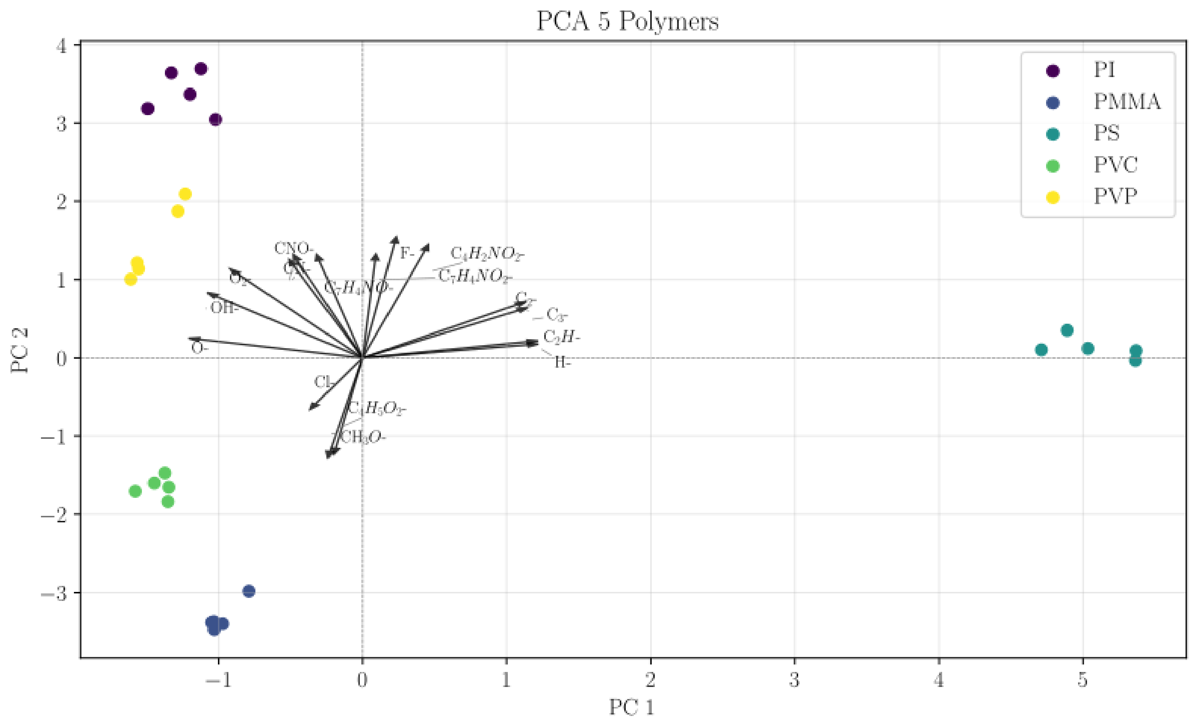


Figure 11: Principal component analysis (PCA) of 5 measurements from 5 different polymers (PVP, PVC, PI, PS, PMMA). Normalized and standardized peak area of selected spectral descriptors. PC1 and PC2 projection including loadings vectors.

7.3 Differentiating Polyimides from Different Sources

To further investigate the feasibility of this chemometric approach, another classification experiment was performed. Three types of polyimides were used, the PI wafer, PI foil and drop cast PI, as their composition is more similar when compared to the selection in 7.2.

As before, 5 repeated measurements of static SIMS were performed per sample and the resulting HCA, Figure 12, again showed clear clustering of the different PI types. Interpretation here is not as intuitive, but samples could be categorized into "commercial" for the foil and wafer and into "non-commercial" for the drop cast PI solution. Principal component analysis (Figure 13) also managed to separate the different types. The drop casted sample shows highest variance, due to being the most inhomogeneous sample compared to the coated wafer and the foil. Loadings are harder to interpret, as the identity of additives in the coated wafer sample is unknown, but the main driving force behind the separation seems to be CN^- and CNO^- in PC1 direction. PC2 is not interpretable in a meaningful way.

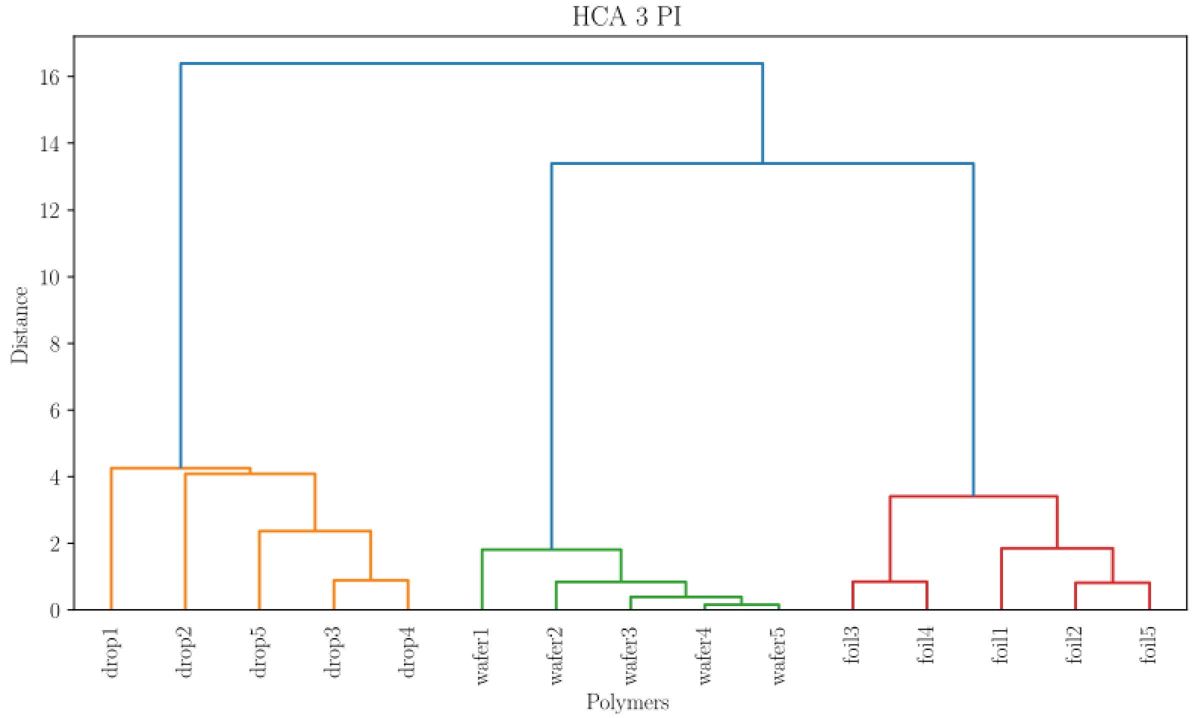


Figure 12: Hierarchical cluster analysis (HCA) of 5 measurements from 3 different polyimides (coated Si-wafer, foil, dissolved powder). Normalized and standardized peak area of selected spectral descriptors, Ward's method, Euclidian distance.

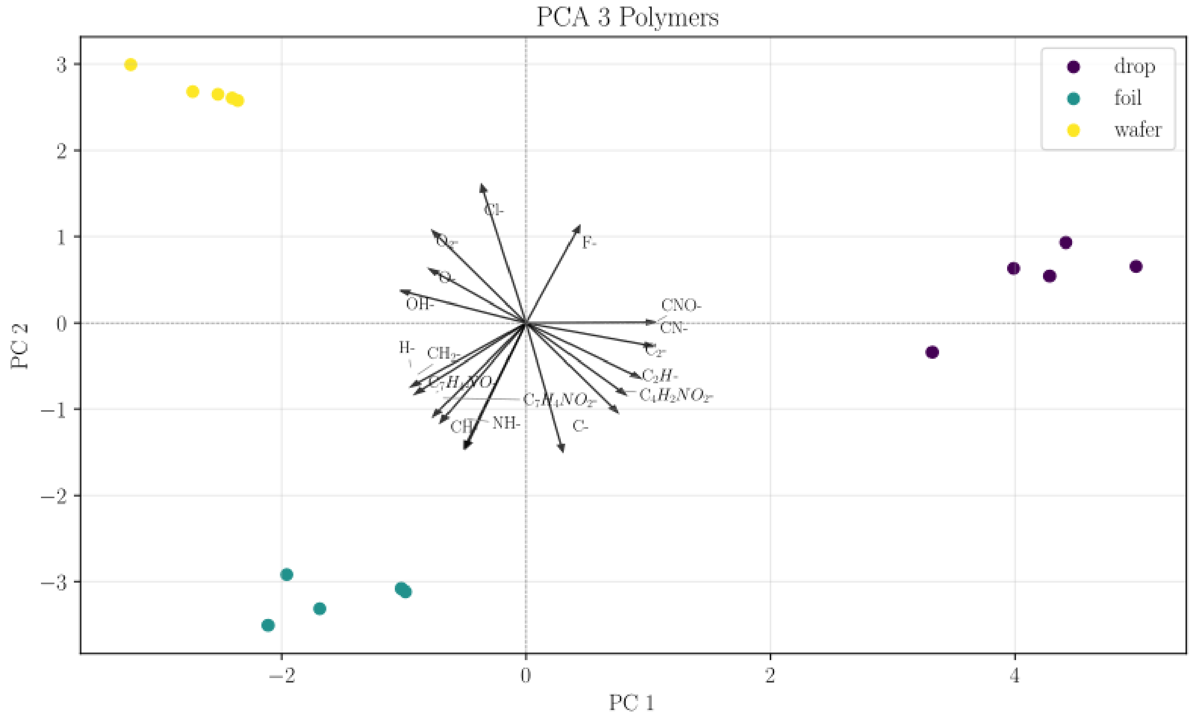
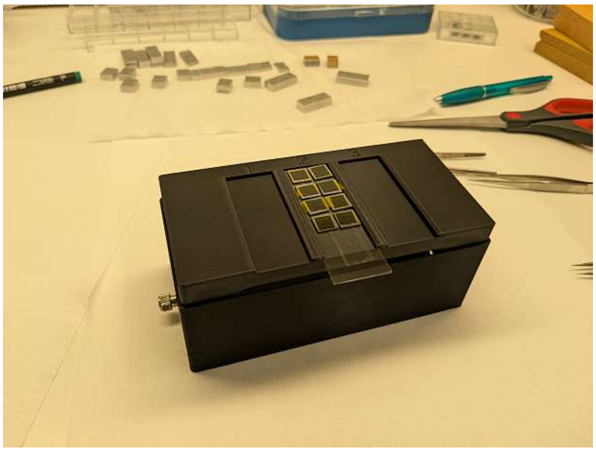


Figure 13: Principal component analysis (PCA) of 5 measurements from 3 different polyimides (coated Si-wafer, foil, dissolved powder). Normalized and standardized peak area of selected spectral descriptors. PC1 and PC2 projection including loadings vectors.

7.4 Aging of Polymer Samples

With the core concept being verified and applicable to the research question, the main experiment now can be performed. The idea is that in an aged polymer, when compared to an unaltered reference sample, high signal variance can be found near the surface of a polymer, which is gradually fading into little to no variance when reaching the unaltered bulk of the material. We hypothesize that the bulk of the aged sample and the bulk of the reference sample are comparable. PCA will be used to investigate the propagation of variance through a depth profile.

The samples used were the PI coated wafer, as well as PBO coated wafers, both used with no further sample preparation. To age samples UV light was used, as it is a simple and quick method of introducing degradation, without introducing foreign elements into the polymer, or contaminating the polymer in any other unpredictable way. A 11 W UV-tube was used in a thin film chromatography UV box. The samples were placed onto glass microscope slides, which were held by a 3D printed PLA (polylactic acid) sample tray, at an approximate distance of 70 mm to the UV tube. Aging times for the samples were 10, 20, 30 and 40 h, while the reference sample was stored in the same hood, on top of the UV box to expose it to the same environment, just without the UV radiation.



(a)

(b)

Figure 14: Sample holder, 3D printed PLA with (a glass microscope slide and samples on top, (b) in the 11 W thin film chromatography UV box.

Depth profiles of all samples were performed with the established settings. A sputter time of 500 s was deemed sufficient, as an initial overview with a sputter time of 1000 s did not show any significant changes beyond 300-400 s for PI, as an equilibrium between ion intensities in the bulk of the material was reached. After mass calibration and selection of spectral descriptors, the exported data was processed using a python script. Crater depth was measured on the DektakXT, taking the average of three passes. One on the left, one at the center and one on the right side of the crater.

7.4.1 Polyimide (PI)

Polyimides are known for their high end properties, including mechanical resistance, outstanding stability under high temperatures, electrical insulation properties and chemical resistance. Although electrical insulation of PI is not keeping up with today's needs, modifications of PI have been developed, improving this property to satisfactory levels. Their modifiable structure is another important characteristic of polyimides. The semiconductor industry uses polyimides, for example Kapton, Figure 15, on the regular as passivation layers and protective coatings, interlayer insulation, and high temperature adhesive.[30]

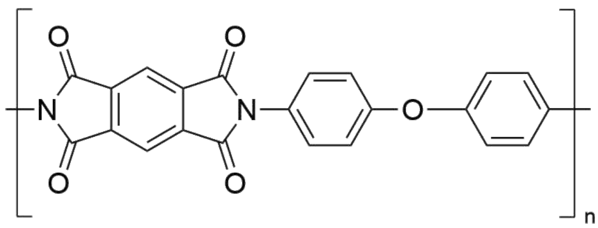


Figure 15: Structure of Kapton, a well known polyimide.

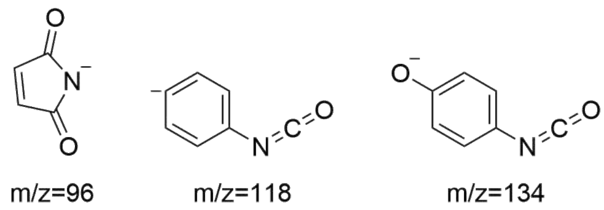


Figure 16: Selected characteristic fragments for polyimide.

Spectral descriptors for PI were chosen, some unspecific and some Figure 16 specific, although technically not needed to perform the analysis. After determining sputter depth with the stylus profilometer, substantially different sputter depths appeared in Figure 17, which is plotted from the values in Table 2, giving first indications on the progression of degradation. The plot indicates a linear connection between aging time and sputter depth.

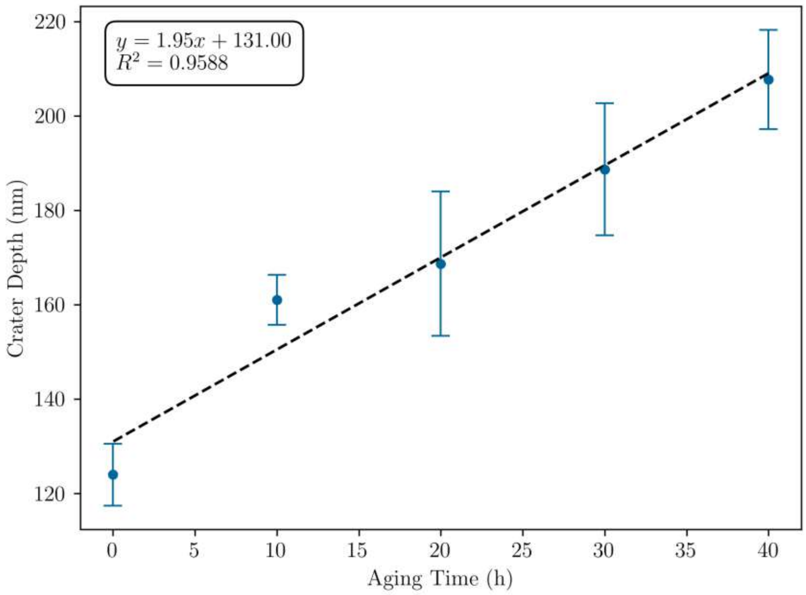


Figure 17: Depth values for ToF-SIMS craters in polyimide film on Si wafer, after 500 s sputtering with 1 keV Cs^+ beam, fitted function indicates matrix changes linearly with sputter time.

Table 2: Depth values for craters in polyimide film on Si wafer, after 500 s sputtering with 1 keV Cs^+, determined with profilometer DektakXT.

Time (h)	Depth (nm)	Stdev (nm)	Sputter Rate (nm/s)
40	207.7	10.5	0.415
30	188.7	14.0	0.377
20	168.7	15.3	0.337
10	161.0	5.3	0.322
0	124.0	6.6	0.248

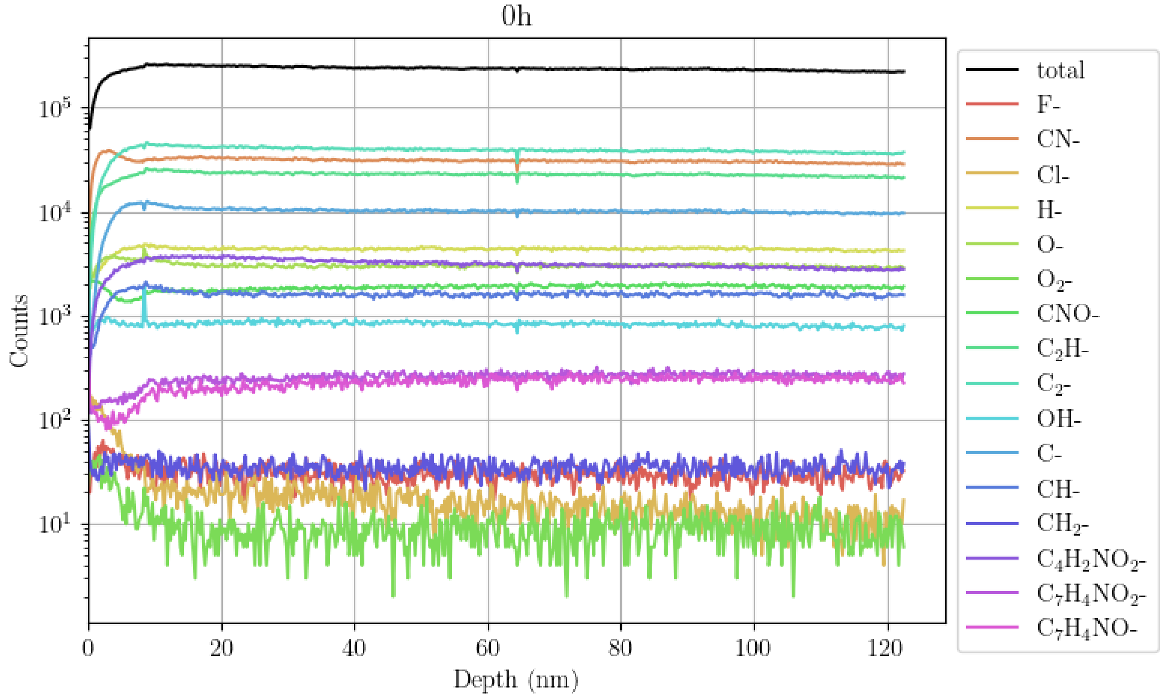


Figure 18: Depth profile polyimide on Si-wafer, without UV aging (0 h), blank, 500 s sputtering 1 keV Cs^+.

The blank measurement in Figure 18 shows equilibration of all chosen species after only a few nm (≈ 20 nm) of sputtering, producing a constant signal over the entire measurement range. The only noticeable variations occur right at the beginning, as steady state sputtering is not fully reached, and possible adsorbates from the surface are removed by the sputter beam. Any spikes in the signal (e.g. 10 nm or 65 nm) are artifacts from a, at the time, unstable primary ion source.

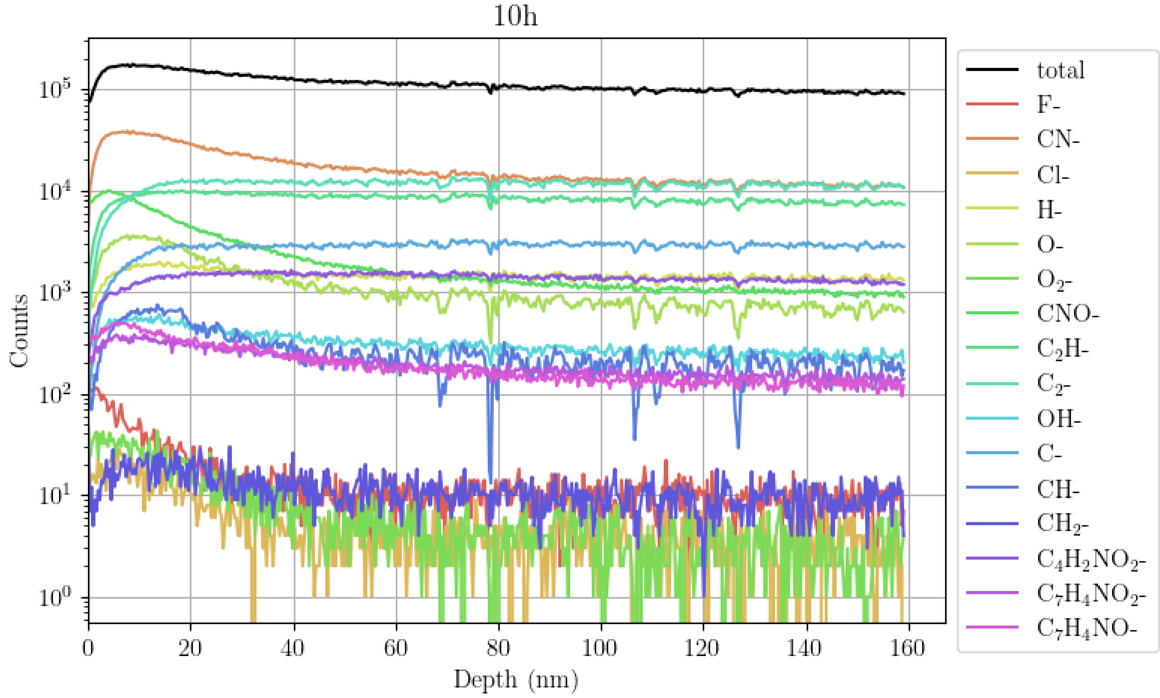


Figure 19: Depth profile polyimide on Si-wafer, UV aged for 10 h, 500 s sputtering 1 keV Cs^+.

After 10 h of UV treatment, significant differences can be observed near the surface(Figure 19). Especially CN^- and CNO^- show a pronounced, slowly fading peak at the beginning of the measurement. Overall counts are higher at the beginning, before flattening into a constant bulk signal, indicating a matrix change in the aged polymer near-surface areas and therefore changed ionization efficiency.

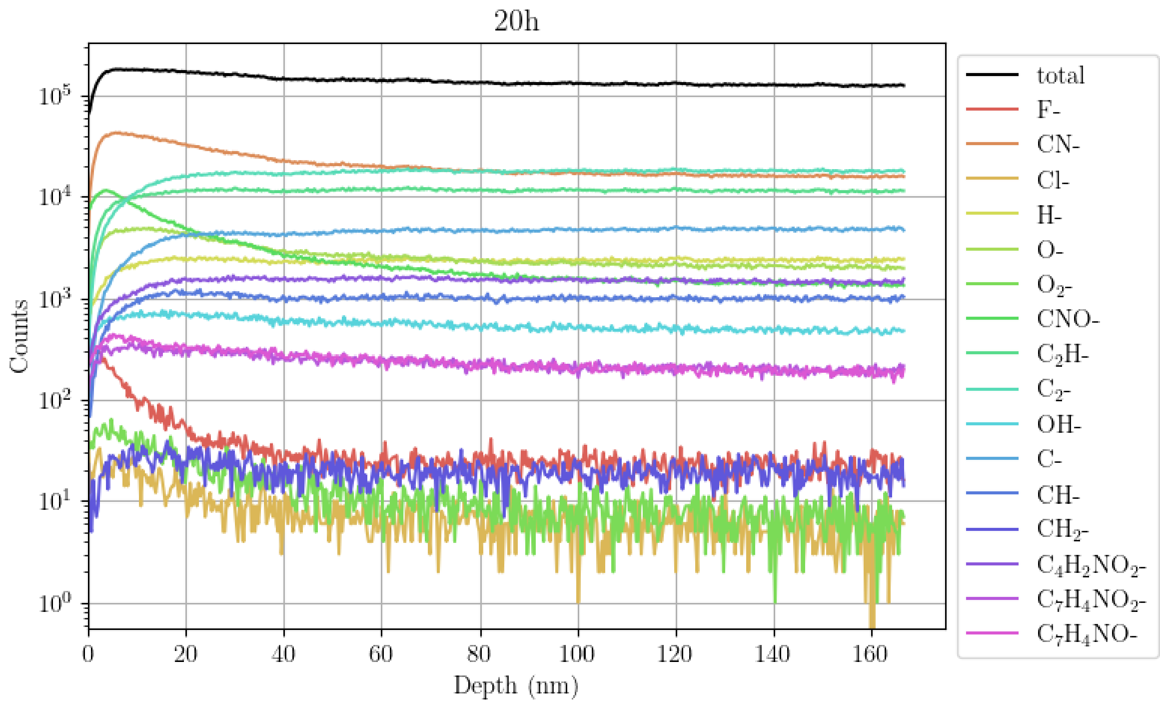


Figure 20: Depth profile polyimide on Si-wafer, UV aged for 20 h, 500 s sputtering 1 keV Cs^+.

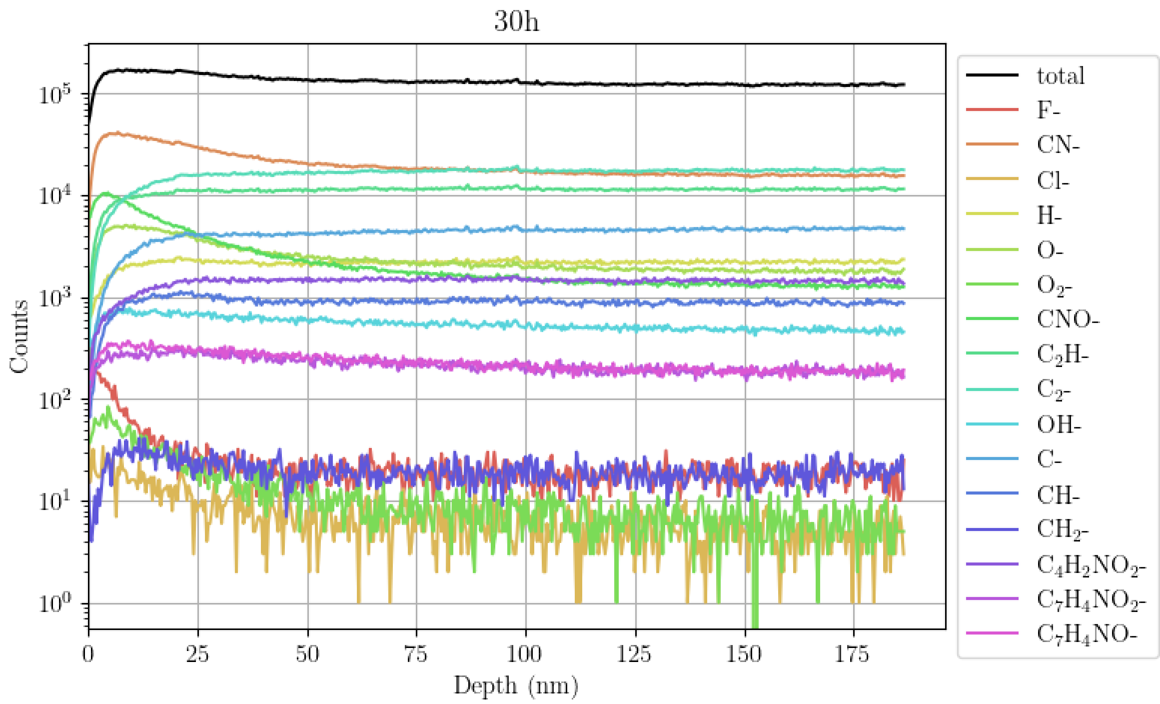


Figure 21: Depth profile polyimide on Si-wafer, UV aged for 30 h, 500 s sputtering 1 keV Cs^+.

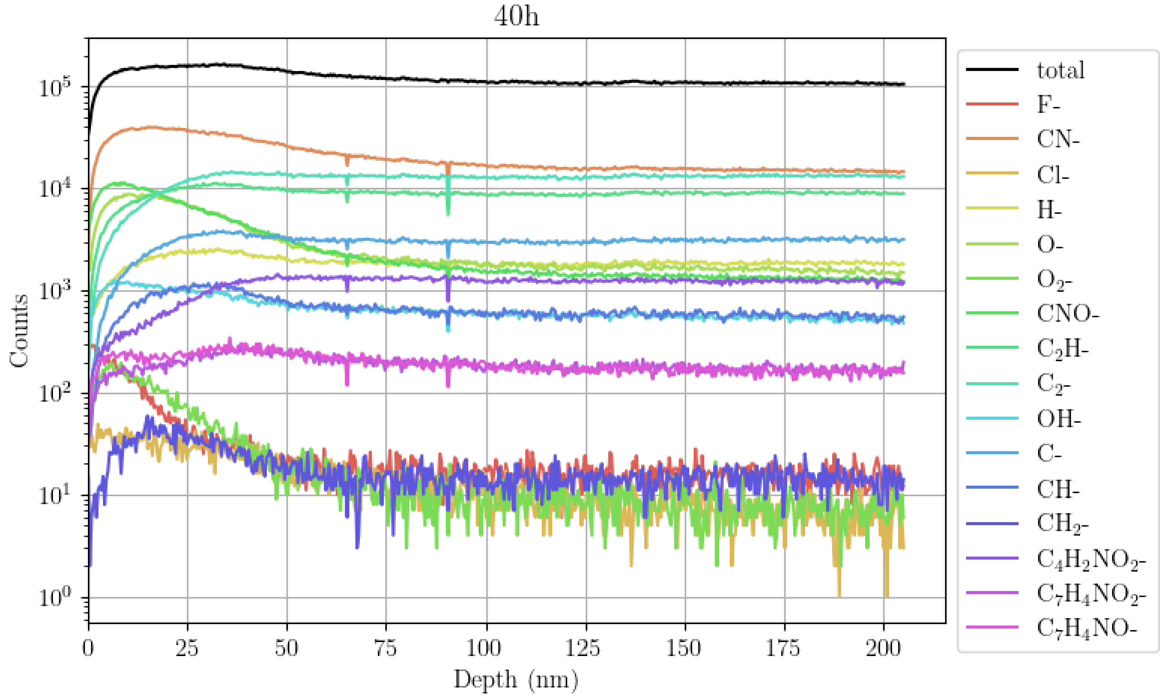


Figure 22: Depth profile polyimide on Si-wafer, UV aged for 40 h, 500 s sputtering 1 keV Cs^+.

In the previous depth profiles Figure 20, Figure 21, Figure 22, differing intensities near the surface can be observed, as steady state sputtering needs to be reached for signals to stabilize. Figure 23 demonstrates that some ion species, like CN^- develop large changes in intensity across the applied aging range, while others, such as OH^- do not. Absolute intensities cannot be directly compared, as they depend on the aging time of the sample. However, the progression or trend of intensities over time can be analyzed, to show differences between aging times. With further aging, the peaks near the surface for CN^- and CNO^- spread outwardly, until they gradually smooth out into the constant intensities of the bulk. Slight qualitative differences can not be reliably ascertained by visual inspection, but need further statistical data analysis. Usually the first 5-10 sputter seconds can be discarded, as adsorbates distort information content near and at the surface, in all depth profiles and PCA plots, the first 10 s have been removed for this reason. Intensities are higher on the aged samples, which can be explained by oxidation and degradation near the surface, leading to a pronounced matrix effect, changing ionization probabilities and therefore sensitivities in a significant way.

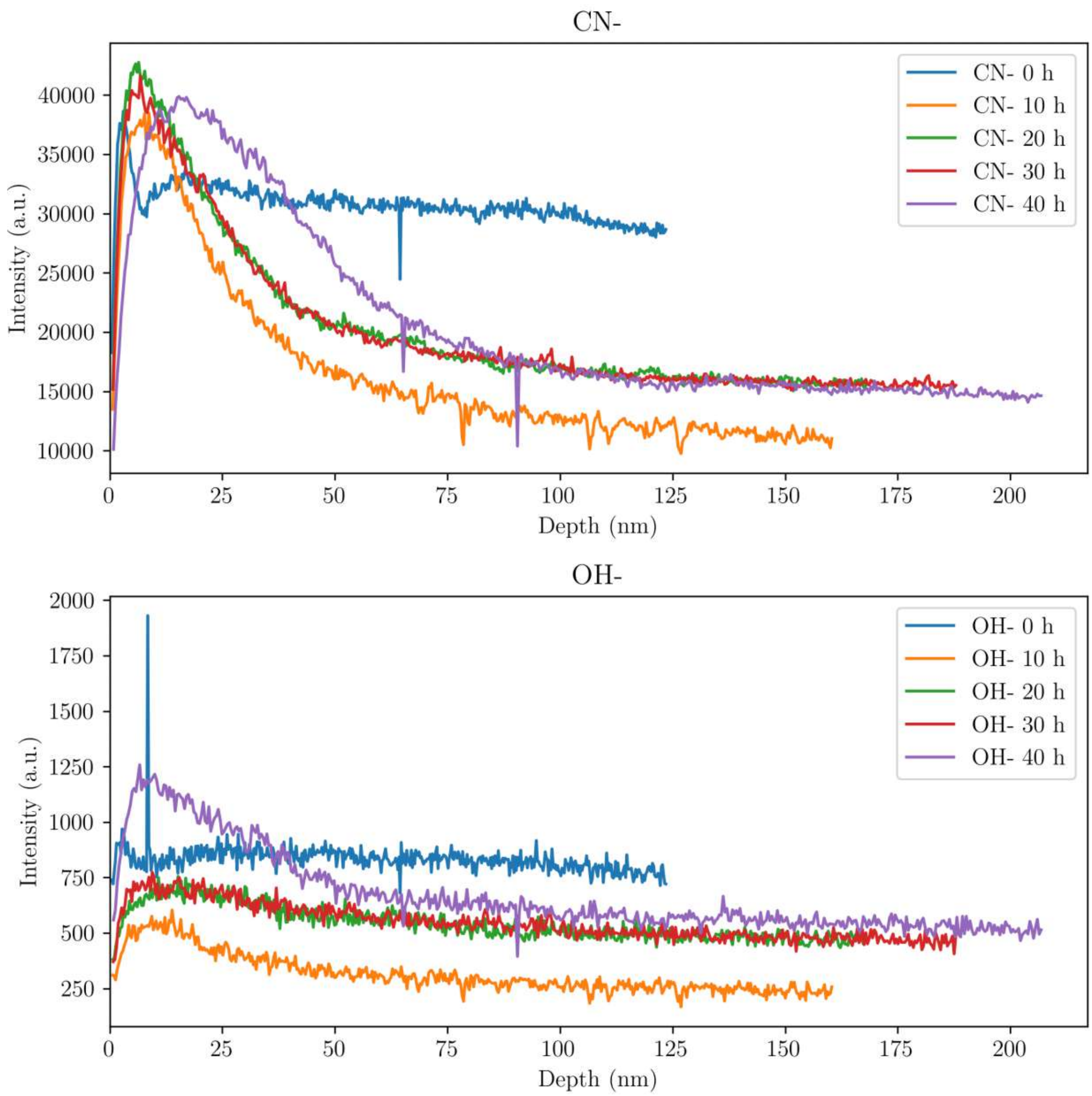


Figure 23: Comparison of ion intensities in PI across aging range. CN^- shows larger changes, OH^- only minute changes.

To compare the aged samples to the blank sample, two depth profile datasets were merged and their ion intensities (row wise standardized and column wise normalized) analyzed using PCA. The same spectral descriptors as in the depth profiles were used to create the PCA plots. According to our hypothesis and the above depth profiles, the variance in the first few nm of the sample should be high, resulting in spread data points in the PCA, as PC1 and PC2 capture the two directions with the highest variance. Approaching bulk depths, variance should decrease, ending in a lower variance cloud of data points. Ideally the bulk data point clouds of the aged and the blank sample would overlap, not only reaching their sputter equilibrium, but also merging with the datapoints of the other sample.

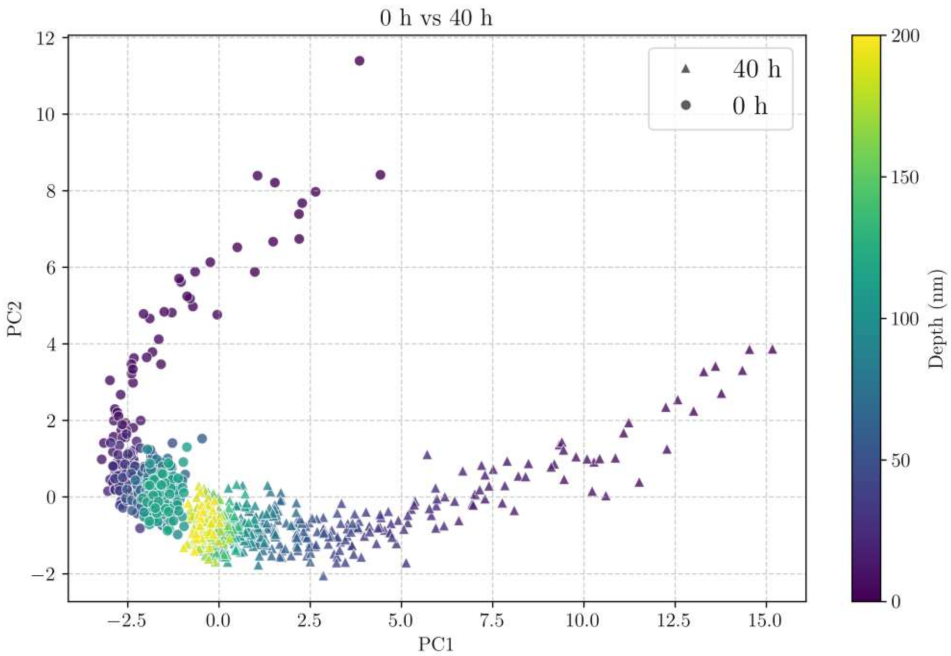


Figure 24: Principal component analysis of UV aged PI sample (40 h) and blank sample, after 500 s of sputtering. Color gradient indicates depth of the data points, showing high variance near the surface, significantly different between aged and blank, merging data points further into the sample when reaching bulk.

In Figure 24 an outcome which corresponds with the initial hypothesis can be seen. High variance is observed near the surface, but different for sample and blank. Both data sets progress into lower variance, and both data clouds converge at higher depth values, indicating that bulk composition in the blank and aged sample are similar.

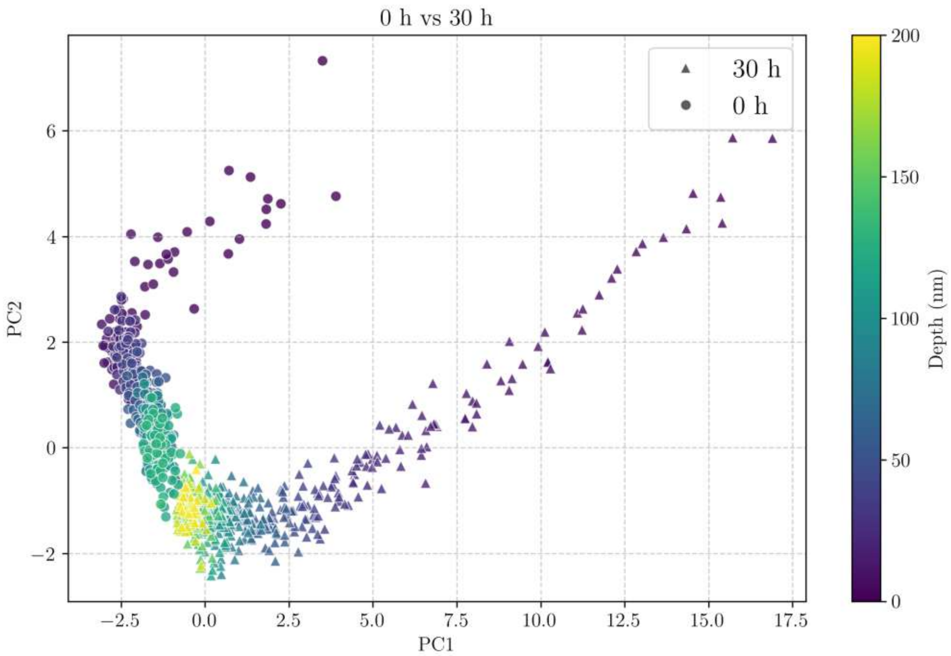


Figure 25: Principal component analysis of UV aged PI sample (30 h) and blank sample, after 500 s of sputtering. Color gradient indicates depth of the data points, showing high variance near the surface, significantly different between aged and blank, merging data points further into the sample when reaching bulk.

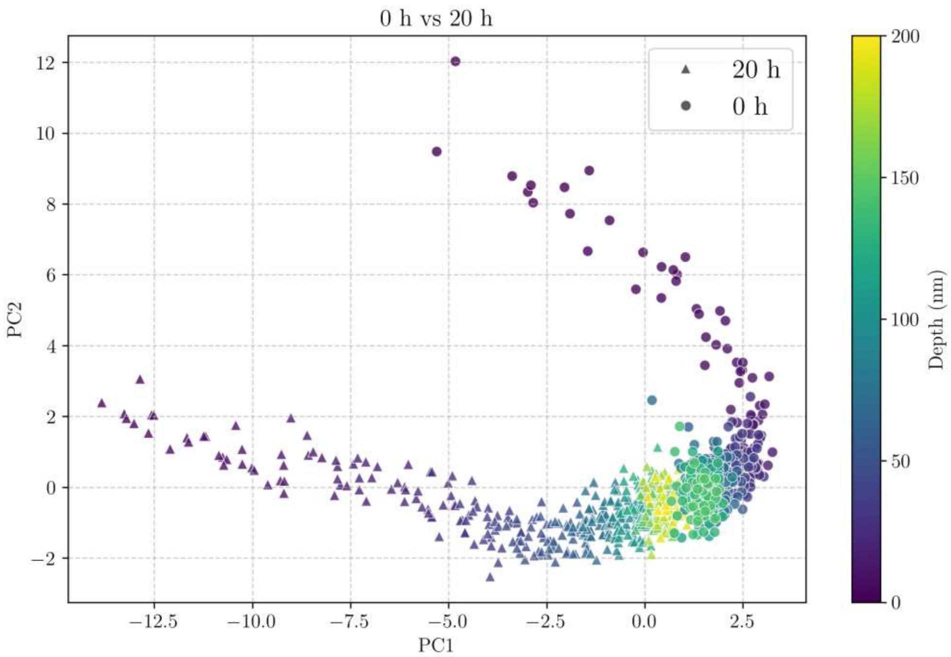


Figure 26: Principal component analysis of UV aged PI sample (20 h) and blank sample, after 500 s of sputtering. Color gradient indicates depth of the data points, showing high variance near the surface, significantly different between aged and blank, merging data points further into the sample when reaching bulk.

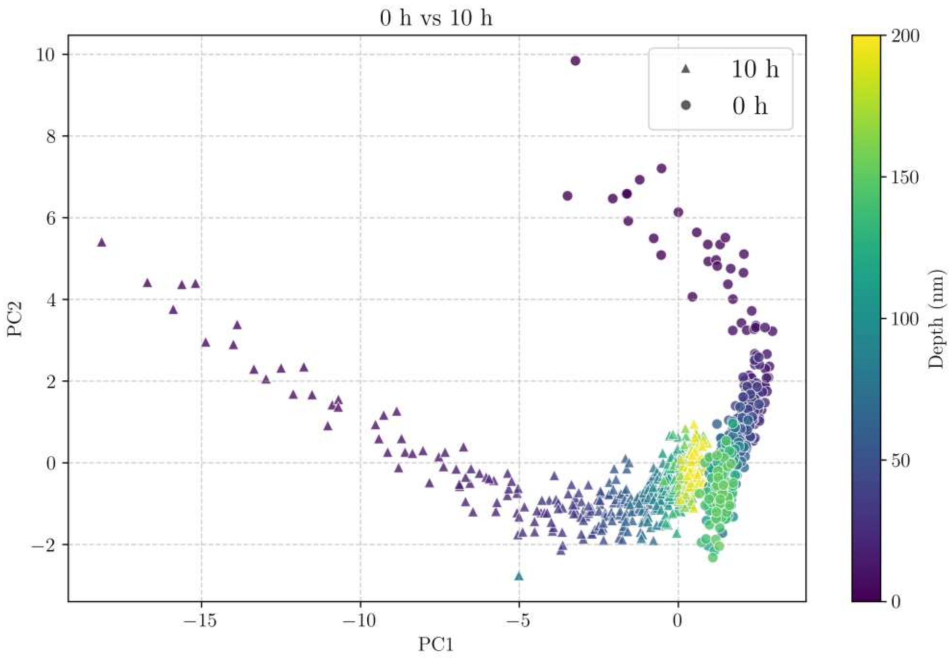


Figure 27: Principal component analysis of UV aged PI sample (10 h) and blank sample, after 500 s of sputtering. Color gradient indicates depth of the data points, showing high variance near the surface, significantly different between aged and blank, merging data points further into the sample when reaching bulk.

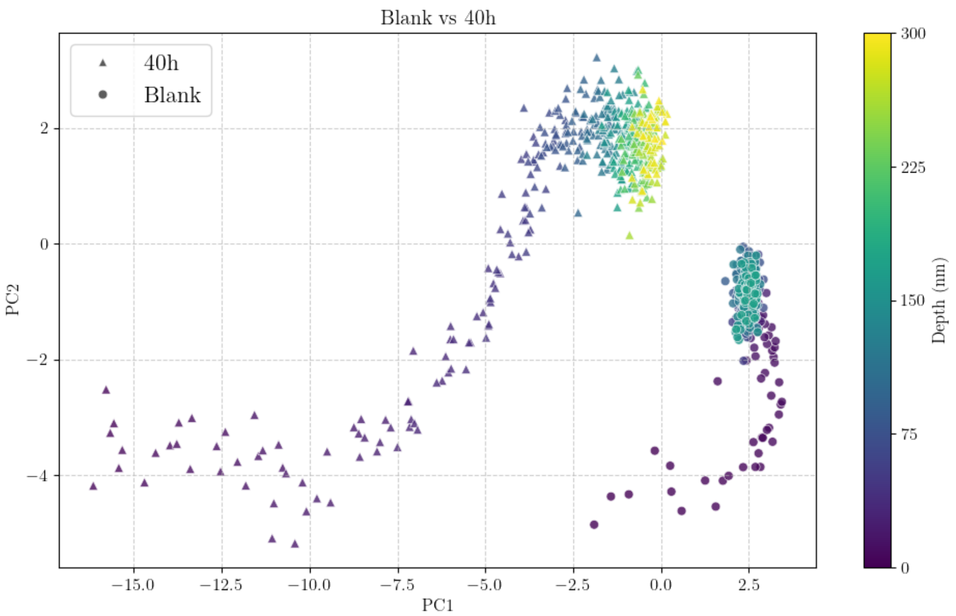


Figure 28: Principal component analysis of UV aged PI sample (40 h) and blank sample from different experiment runs, after 500 s of sputtering. Color gradient indicates depth of the data points, showing high variance near the surface, significantly different between aged and blank, not merging data points further into the sample when reaching bulk.

In general, the data clouds merged or nearly merged upon reaching the bulk material, with the blank sample consistently reaching equilibrium first, as indicated by the depth profiles. However, when comparing measurements from different experimental batches acquired on different days, although normalized to improve comparability, the comparison does not yield the expected results, with day-to-day instrument variation, like vacuum conditions, most likely being the issue. Figure 28 is an example of two datasets not merging at the bulk level, due to being recorded at different days, nevertheless some of the initial hypotheses are supported. Again high variance can be seen near the surface of the sample, with a distinct difference between both samples. Data points in both samples start to move towards the other sample, individually reaching equilibrium, but not merging. The PCA's loadings don't give any indication on why the data points in Figure 24 merge, but not in Figure 28.

7.4.2 poly(p-phenylene-2,6-benzobisoxazole), PBO

PBO is a high performance rigid-rod polymer fiber, developed in the 1970s-1980s, showing extreme thermal stability, tensile strength and creep resistance. Dissolving PBO is not possible, as it is resistant to most solvents and strong acids. PBO is commonly known under its commercial name Zylon and is used as fibers for example in body armors.[31]

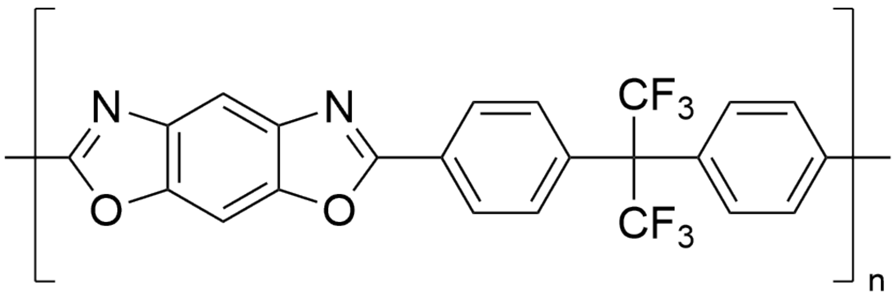


Figure 29: poly(p-phenylene-2,6-benzobisoxazole) PBO, modified with trifluoromethyl groups, (6FPBO), as described by Zhang et al.[32]

For PBO experiments aging times were spaced more narrowly, as preliminary experiments showed that 40 h aging resulted in far more degradation than in PI, which would require exorbitantly longer measuring times, equaling deeper probing of the material. An overview of of the sputter rates plotted over time, Figure 30 with values from Table 3, shows that with longer UV exposure the sputter depth after 500 s sputtering rises from 160 nm for the blank sample to over 500 nm for the longer aged samples, flattening out with higher aging time. The asymptotic course of the sputtering depth indicates a saturation in degradation near the surface, with the front of degradation moving further into the bulk of the polymer. 5 h aging intervals were chosen, to facilitate the detection of the time, when degradation starts to slow down. Compared to the PI craters, PBO seems to be more readily and consistently sputtered, giving less deviation as PI, Figure 17.

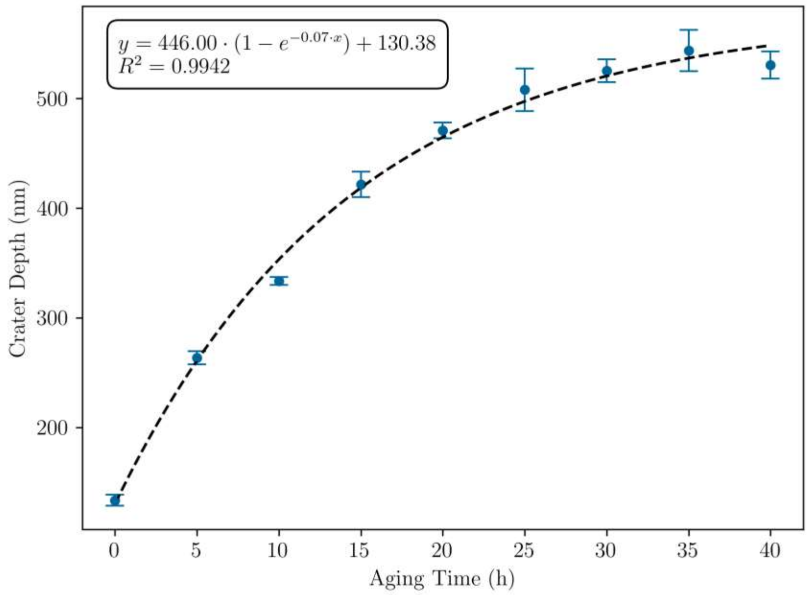


Figure 30: Depth values for SIMS craters in poly(p-phenylene-2,6-benzobisoxazole) PBO, after 500 s sputtering with 1 keV Cs beam. The fitted curve indicates matrix changes slow down with longer aging times and indicates an asymptotic behavior.

Table 3: Depth values for craters in poly(p-phenylene-2,6-benzobisoxazole) (PBO) after 500 s 1 keV Cs sputtering, determined with profilometer DekTak XT.

Time (h)	Depth (nm)	Stdev (nm)	Sputter Rate (nm/s)
40	530.7	12.3	1.061
35	543.7	18.9	1.087
30	525.3	10.5	1.051
25	508.0	19.5	1.016
20	471.0	7.1	0.942
15	421.7	11.5	0.843
10	333.7	3.5	0.667
5	263.7	6.1	0.527
0	133.7	4.9	0.267

After aging, the samples were also measured with ToF-SIMS settings analogous to the polyimide samples. Initial measurements showed very high F^- response. As the exact composition of the polymer and it's additives were not provided, an assumption was made, that the monomer backbone of the polymer could look similar to Figure 29, as this modification with trifluoromethyl has been reported,[32] and would explain the high F^- content. These modifications in high κ polyimides can lead to better polymer solubility.[33] Spectral descriptors were chosen according to Figure 31, being mostly equivalent to the PI experiments, again the same descriptors were used in the depth profiles and in the PCA plots. Large characteristic fragments, as used in PI, were not used in this experiment, only smaller fragments containing fluorine were chose, due to their abundance in initial measurements. The same data analysis as used for the PI measurements was applied to the depth profiles Figure 32-Figure 37, in order to highlight not only the differences but also the similarities in degradation between the two different polymer types. Equilibrium between the ion intensities in the blank sample is achieved after approximately 60 nm, taking longer than in PI.

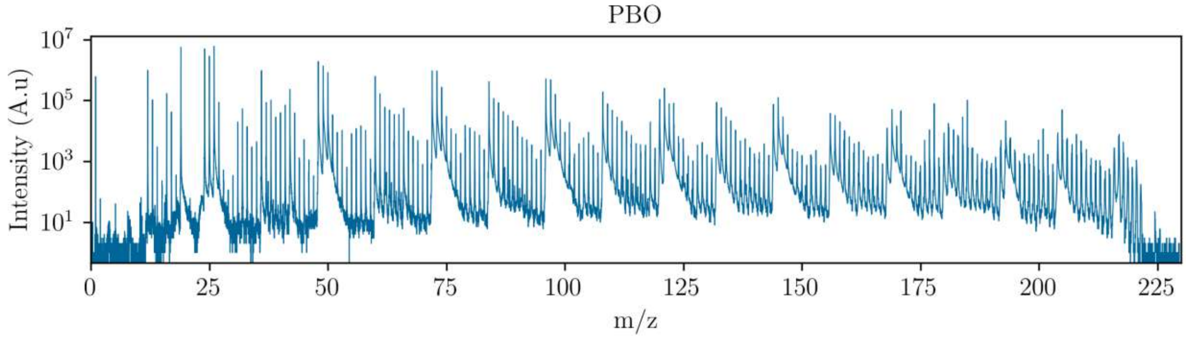


Figure 31: Mass spectrum of PBO, standard settings from section 7.1.6.

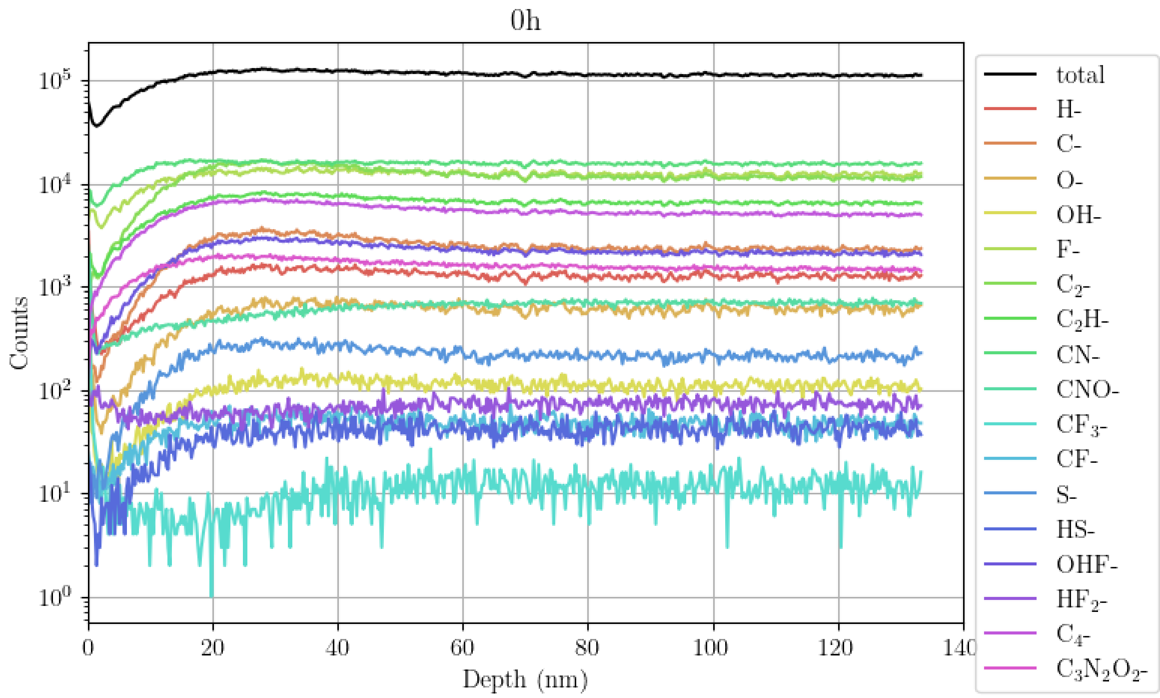


Figure 32: Depth profile PBO on Si-wafer, no UV aging, 500 s sputtering 1 keV Cs^+.

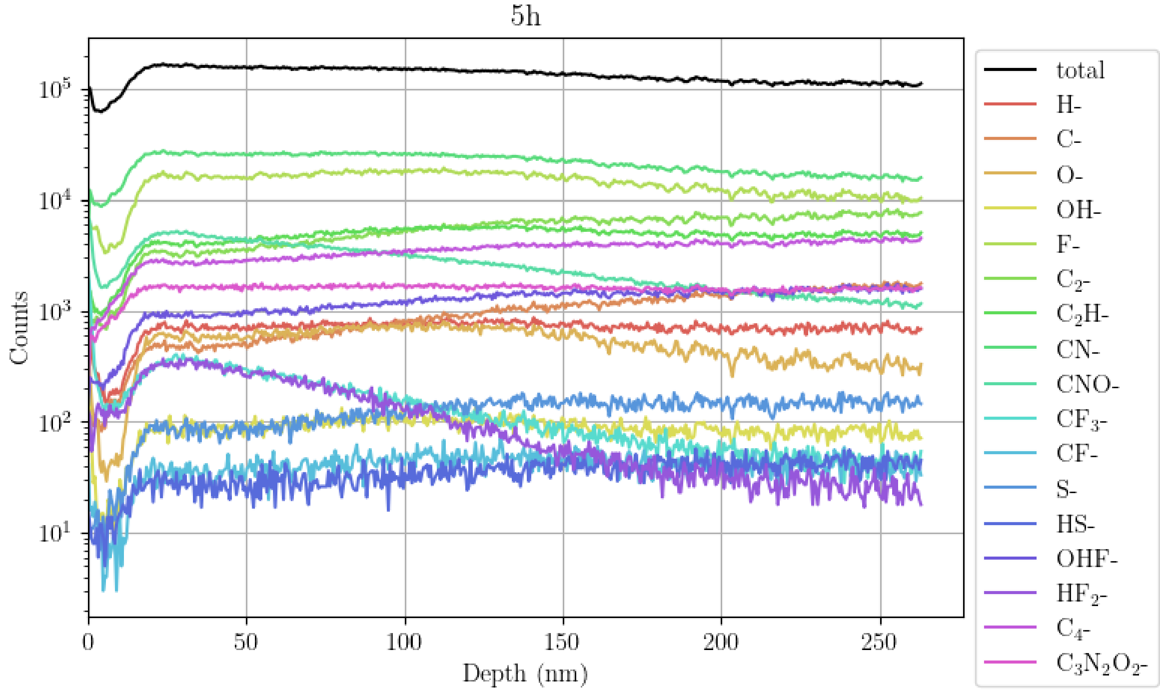


Figure 33: Depth profile PBO on Si-wafer, 5 h UV aging, 500 s sputtering 1 keV Cs^+.

The following Figures show increasing stages of degradation, differing mostly in sputter depth, and the extent of some affected peaks reaching further into the material. Ions like C^- and O^- loose intensity near the surface with longer aging times, again showing a significant change in the material.

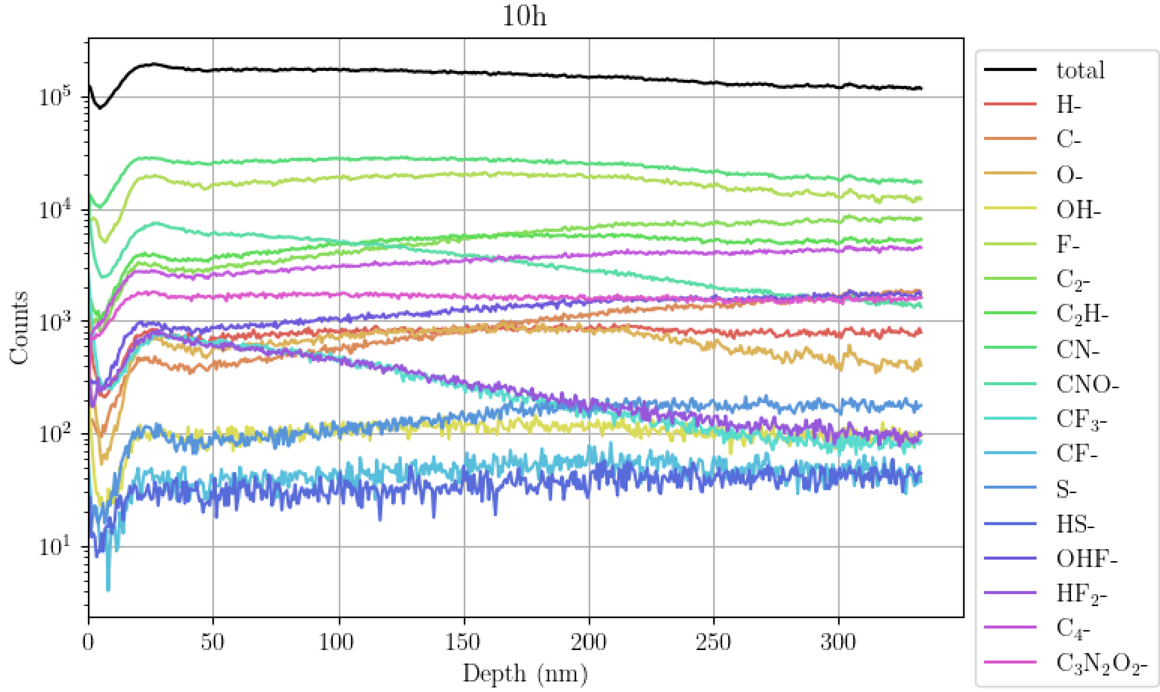


Figure 34: Depth profile PBO on Si-wafer, 10 h UV aging, 500 s sputtering 1 keV Cs^+.

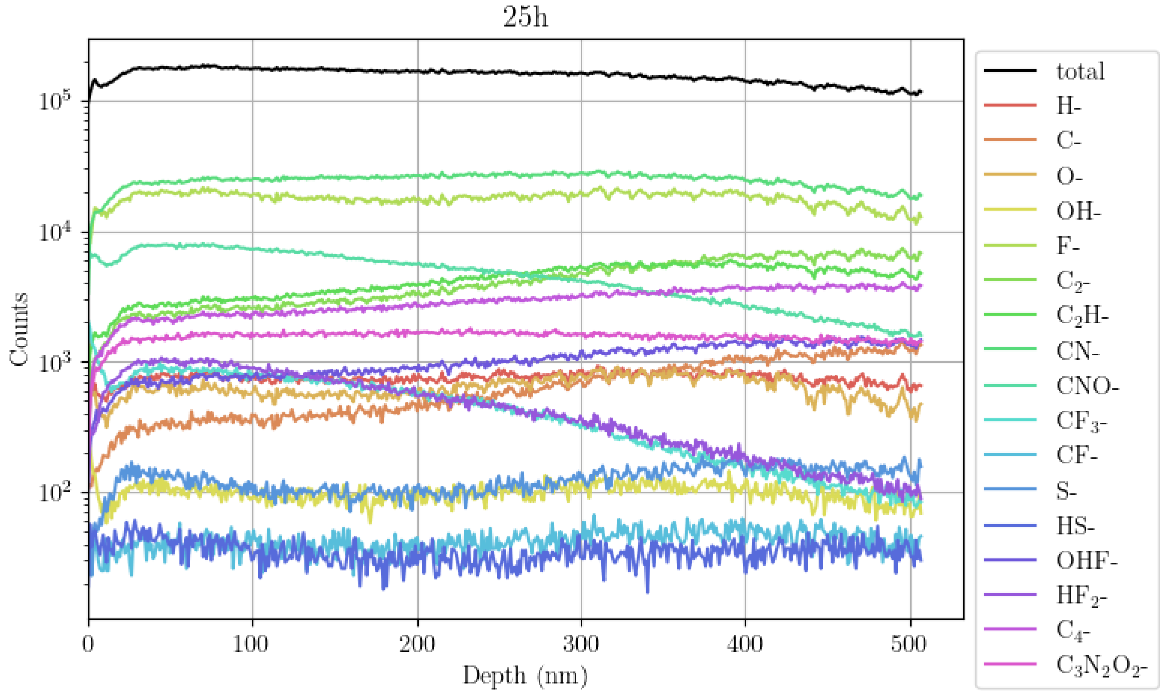


Figure 35: Depth profile PBO on Si-wafer, 25 h UV aging, 500 s sputtering 1 keV Cs^+.

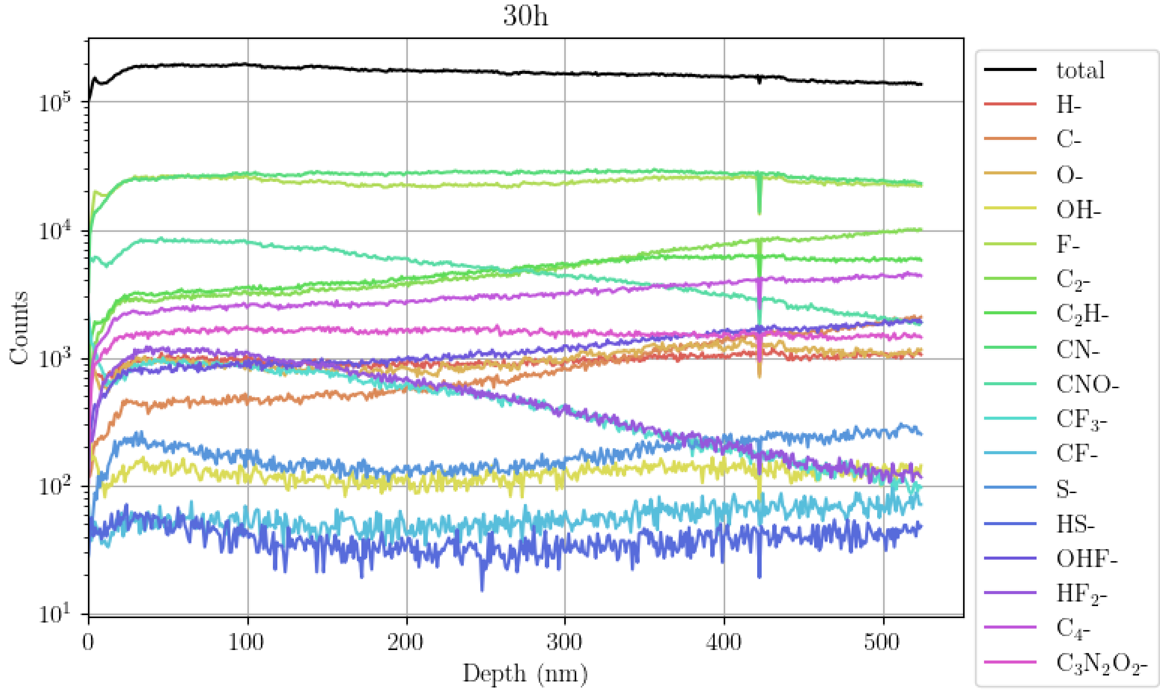


Figure 36: Depth profile PBO on Si-wafer, 30 h UV aging, 500 s sputtering 1 keV Cs^+, signal spikes at around 420 nm.

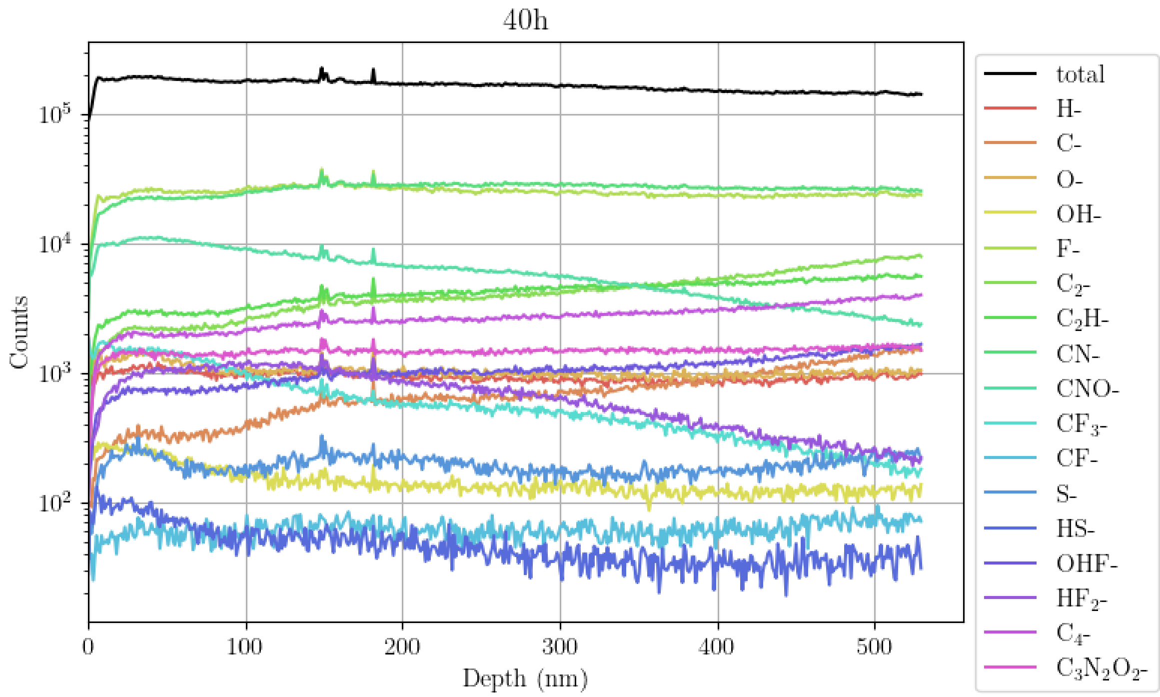


Figure 37: Depth profile PBO on Si-wafer, 40 h UV aging, 500 s sputtering 1 keV Cs^+.

As already indicated by the graph in Figure 30, Figure 37 shows a plateau in the degradation peak of HF_2^-, from 0 nm to almost 500 nm, suggesting saturation of the matrix changes near the surface that impact signal generation in ToF-SIMS measurements. Some ion intensities, such as that of OH^-, do not appear to be affected much by aging, as their intensity profiles remain largely unchanged across different aging times, apart from a shift due to increased sputtering depth. Most affected are some initially less abundant fluorine species, such as CF_3^- and HF_2^-, whose intensities can increase drastically due to matrix changes triggered by the aging process, with their abundance moving further into the polymer with longer aging times. Additionally CF_3^- is an indicator for the proposed structure of PBO in Figure 29.

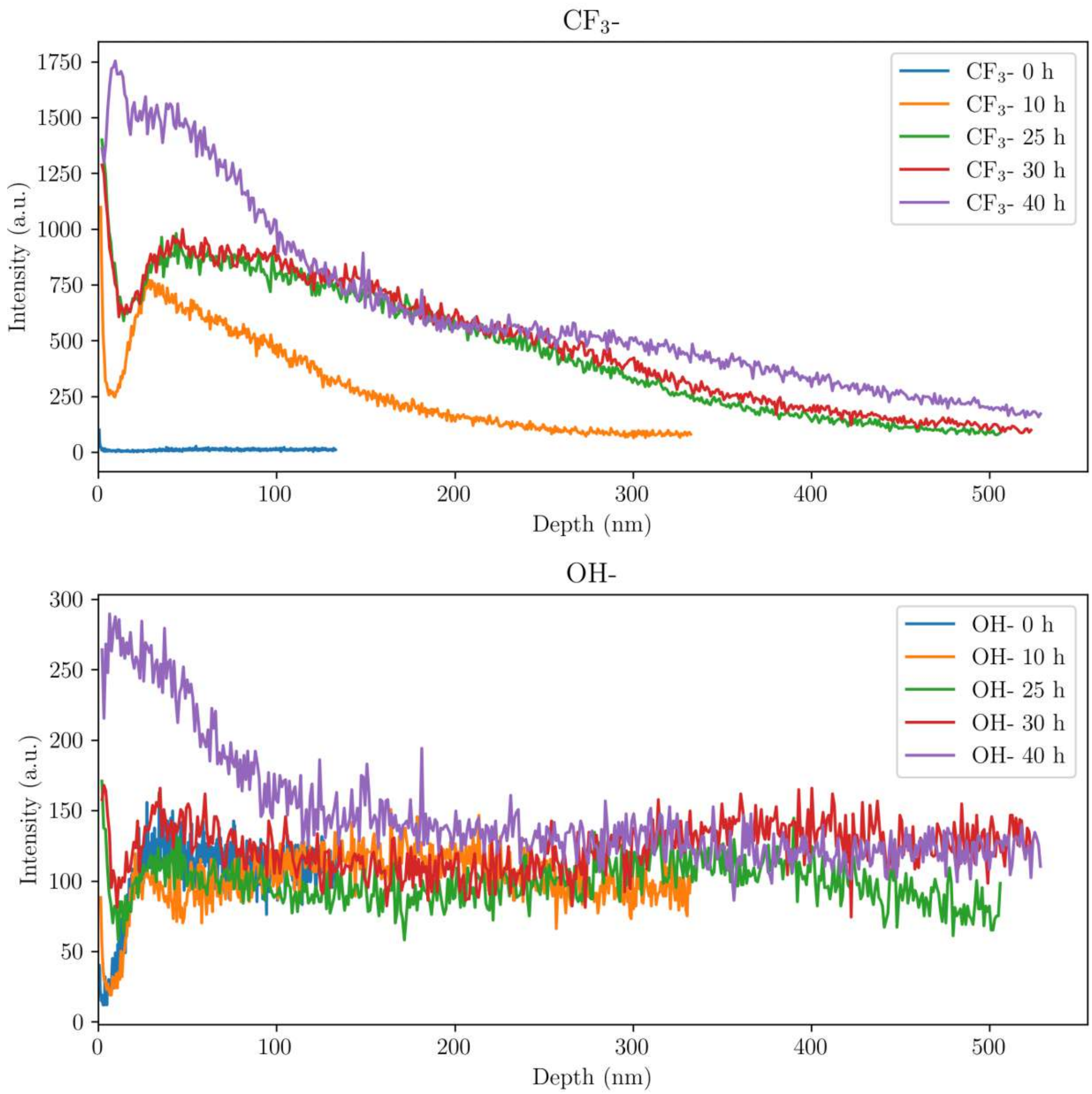


Figure 38: Comparison of ion intensities in PBO across aging range. CF_3^- shows larger changes, OH^- only minute changes.

As seen with the comparisons over the aging times of PI, PBO shows similar behavior in Figure 38. Some ions like CF_3^- change greatly over the aging times, while others, again

OH^-, remain relatively stable. With the exception of the sample aged 40 h, OH^- remains nearly at constant intensity over the entire sputter range.

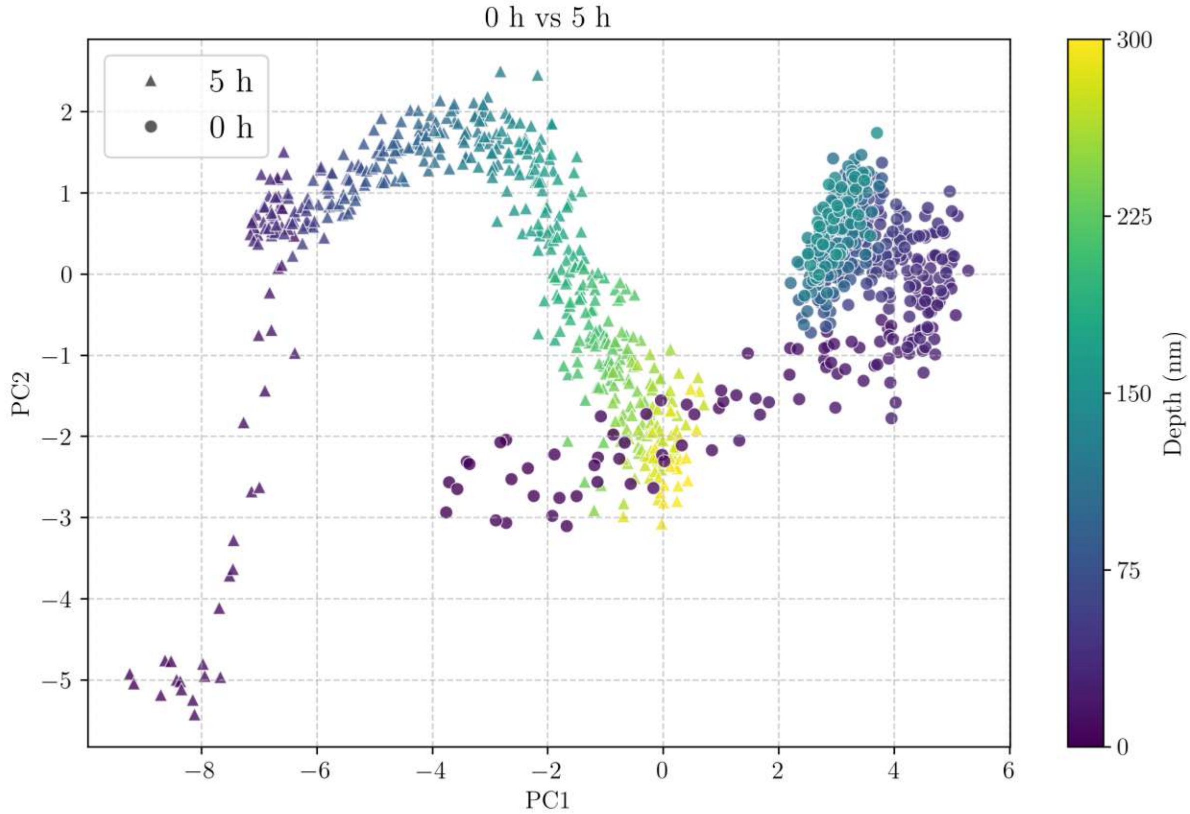


Figure 39: Principal component analysis of UV aged PBO sample (5 h) and blank sample, after 500 s of sputtering. Color gradient indicates depth of the data points, showing high variance near the surface, significantly different between aged and blank.

The PBO PCA plots indicate, that PBO requires much less UV exposure to degrade and change significantly, as PBO is highly absorbant in the UV range used in the aging experiment, with the main absorption peaks located at 404 and 428 nm.[31] The double PCA plots of 0 h and 5 h show that the 0 h sample rapidly reaches equilibrium after around 50 nm, while the 5 h sample already shows high variance over the entire depth profile. Further support for this comes from the measured crater depths, showing a doubling in sputter depth from 0 to 5 h.

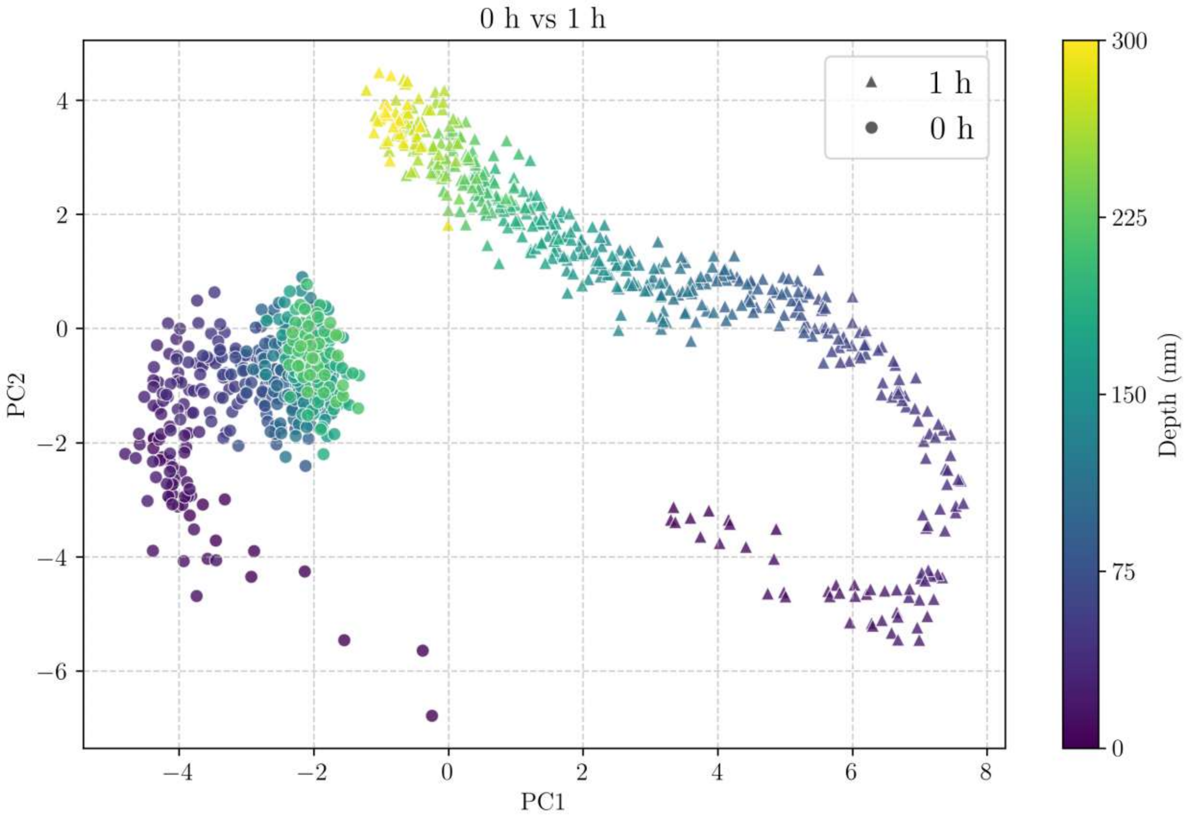


Figure 40: Principal component analysis of UV aged PBO sample (1 h) and blank sample, after 500 s of sputtering.

After just 1h of aging, both samples already show significant differences. However, as the depth profile in Figure 33 indicates, no signal equilibrium is reached within the 500 s of sputtering, suggesting that degradation extends beyond the crater depth of 259 nm.

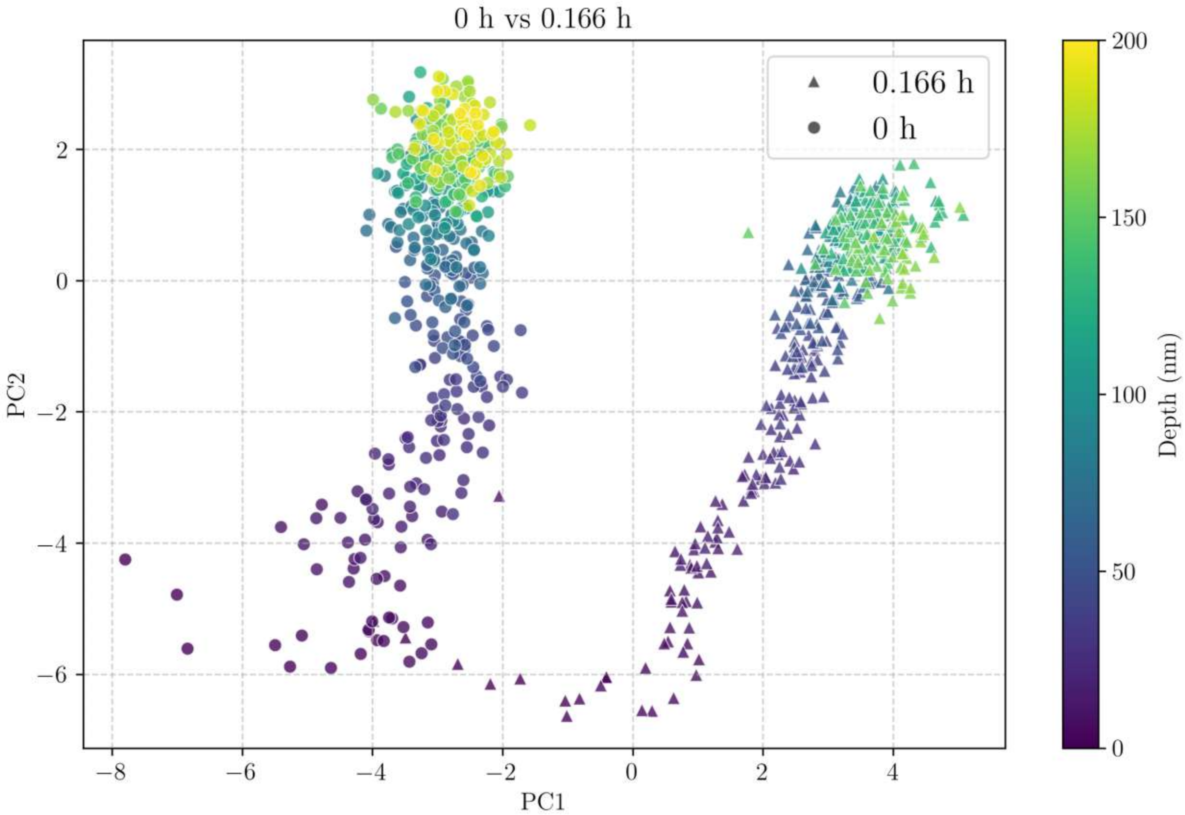


Figure 41: Principal component analysis of UV aged PBO sample (10 min) and blank sample, after 500 s of sputtering.

To confirm whether degradation after 1 h of aging exceeded the sputtered depth reached after 500 s, a 10 min aging time was selected to test if signal equilibrium could be achieved. Aging for 10 min again confirms PBOs susceptibility to UV radiation, as both samples show different behavior at the very beginning of sputtering, and do not merge when reaching bulk. The data points of the aged sample begin to form a cloud with reduced variance at around 100 nm, in contrast to Figure 40 which does not exhibit this effect, indicating that the maximum resolvable aging time with 500 s lies somewhere between 10 min and 1 h.

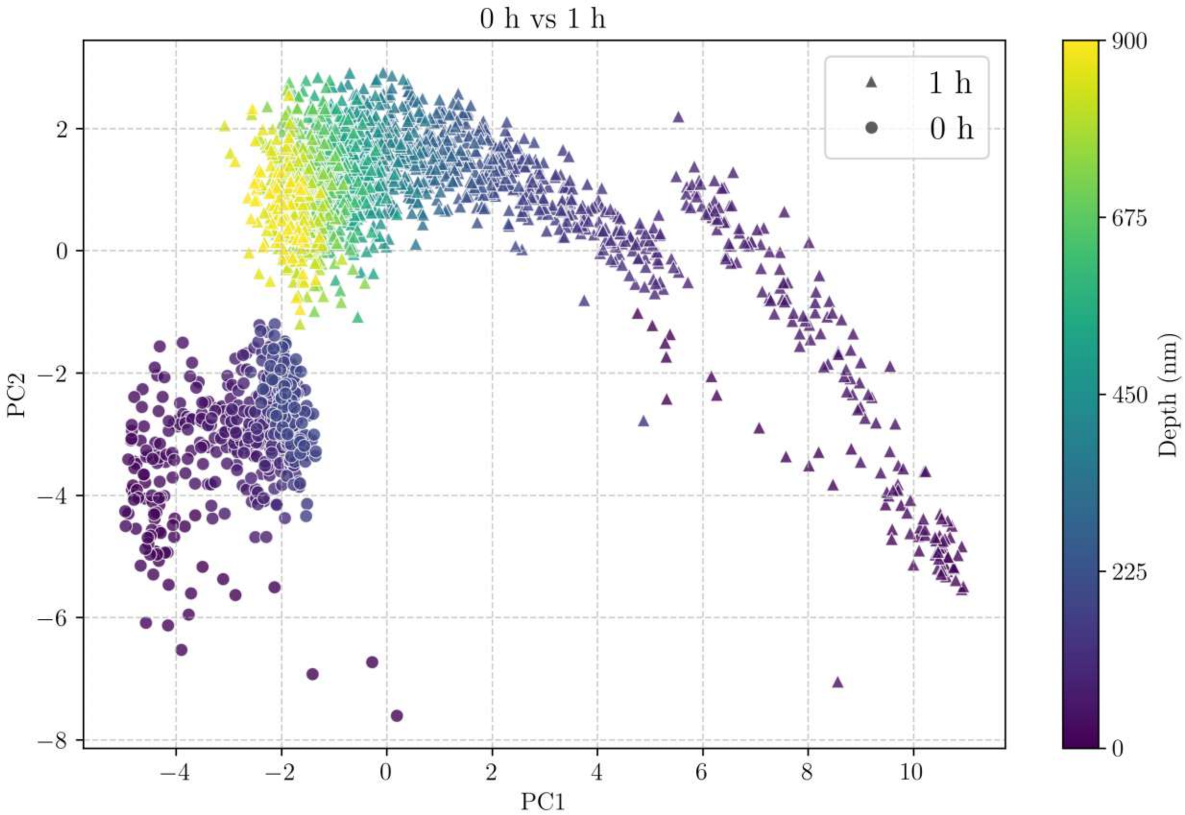


Figure 42: Principal component analysis of UV aged PBO sample (1h) and blank sample, after 1750 s of sputtering for the aged sample and 500 s sputtering for the blank sample.

Extending the sputtering duration, therefore deepening the sputter crater, to 1750 s in this case, reveals that signal equilibrium indeed is reached at a depth of approximately 450 nm, far exceeding the range of 500 s sputtering. Still the datapoint clouds do not overlap between the aged and the native sample and only approach each other. Further investigation would be needed to confirm merging of data clouds, however as a single measurement with 1750 s of sputtering under the current settings already requires 82 min, alternative analysis methods would need to be employed with the ability of more time efficient depth profiling in the upper nanometer to lower micrometer range.

8 Conclusion

Adequate parameters for polymer measurements on the IONTOF ToF-SIMS 5 were established. Bi_3^+ was chosen for the primary species, as it offers a valid compromise between signal intensity and plentiful fragments compared to Bi_1^+ and Bi_5^+. Cycle time was set to 50 µs as fragments beyond 230 u can not be elucidated and are not needed for the chosen data evaluation. 1 keV Cs^+ in negative mode was used as the sputter species, as it has the most favorable fragments, showing a repeating pattern every 12 u, thus, enables sensitive detection of F^- and $Cl^-$and has 100 times higher signal. The floodgun was deemed very important for charge compensation and was run at standard settings. To not saturate the detector the pulse width of the primary beam was changed, to accommodate differing signal intensities from various polymers or aging conditions.

Classification of different polymer types was successfully performed, demonstrating the validity of the optimized ToF-SIMS settings. Continued classification of different polyimide types was also performed successfully, laying the groundwork for differentiating native and aged polymers.

The UV aging experiments showed varying stages of degradation near the surface of polymer films, supporting the hypothesis of high signal variance near the surface and low signal variance after passing through the degraded areas into the bulk of the material. Additionally differences in the near surface areas were visualized using PCA, showing conclusive differences between native polymer and polymer that has aged under UV. Another high end polymer film, PBO, was also exposed to UV radiation, showing clear differences even after 10 min of irradiation. However this system did not show the same outcome as PI, with most parts of the initial hypothesis still being supported by this sample. In retrospect PBO was not an ideal sample to be treated with UV radiation, due to its susceptibility to UV. Importantly, it became evident, that during the experiments, that measurements are only comparable when performed on the same day, as instrument conditions, like vacuum, can shift, and influence overall comparability.

Data processing was conducted in a way that ensures consistent evaluation across different samples, spectral descriptors, and degradation mechanisms, ultimately resulting in interpretable double PCA plots. This approach aims to visualize and investigate near-surface alterations in polymers induced by various industrial processes, and to assess the effectiveness of different polymer types, curing methods, or surface modifications in resisting relevant degradation mechanisms throughout the process chain. Further improvements to the method include identifying and characterizing the degradation products to better understand the underlying mechanisms and enable the selection of suitable materials for a given application.

List of Figures

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References

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